AIDCF

AI Data Centre Facilities Assurance Framework

INFRASTRUCTURE ONTOLOGY & TAXONOMY

Version 1.0

Canonical semantics for capability-centred AI infrastructure assurance

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This document defines the canonical entity classes, taxonomies, relationships, properties, constraints and machine-readable expectations used to represent AI infrastructure within AIDCF.

Document control

Field	Value
Title	AIDCF Infrastructure Ontology & Taxonomy v1.0
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Depends on	AIDCF Manifesto v1.0; AIDCF Core Conceptual Model v1.0
Downstream users	Threat, Hazard & Failure Model; AIDCF-AI Trust Graph Integration Standard; Domain Guides; Master Control Library; Evidence Model; Assessment Methodology; Scoring and Maturity Models; compatible tooling
Normative language	Shall, shall not, should, may and can have the meanings defined in AIDCF Core Conceptual Model v1.0
Intended audience	AI infrastructure architects; data-centre engineers; assessors; operators; owners; technology teams; resilience, security and assurance practitioners; ontology engineers; product teams
Publication boundary	Independent methodology. Not a substitute for law, licensed engineering, safety approval, regulatory certification or vendor design authority.

Executive thesis

AIDCF represents an AI data centre as a capability-supporting system rather than a collection of equipment. The ontology begins with a defined Capability Instance and Service Outcomes, traces them to infrastructure functions and assets, expresses direct, indirect and shared dependencies, records capacity and operating conditions, and provides stable attachment points for failures, safeguards, evidence and assurance decisions.

Canonical chain

Capability Instance → Service Outcome → Required Function → Infrastructure System → Asset Instance → Dependency → Operating Condition → Failure Exposure → Safeguard → Evidence → Assurance Decision

How to use this standard

- Use Parts I-IV to establish a conformant assessment information model.
- Use Parts V-VII to classify infrastructure across the eight AIDCF domains.
- Use Parts VIII-XI to represent relationships, dependencies, states, capacity, responsibility and uncertainty.
- Use Parts XII-XIV to validate records, create implementation profiles and produce machine-readable releases.
- Do not treat the taxonomy as a control checklist or a substitute for engineering analysis.

Contents

1. Part I - Authority, scope and conformance
2. Part II - Ontology architecture and design principles
3. Part III - Upper ontology and canonical entities
4. Part IV - Capability, outcome, boundary and lifecycle model
5. Part V - Eight-domain infrastructure taxonomy
6. Part VI - Cross-cutting classification facets
7. Part VII - Asset, system, interface and resource model
8. Part VIII - Relationship and dependency vocabulary
9. Part IX - Capacity, redundancy, fault-domain and resilience semantics
10. Part X - Operating state, temporal state and change semantics
11. Part XI - Responsibility, evidence, confidence and UNKNOWN semantics
12. Part XII - Conformance rules and validation shapes
13. Part XIII - Reference implementation profiles
14. Part XIV - Machine-readable publication specification
15. Appendix A - Canonical class registry
16. Appendix B - Canonical relationship registry
17. Appendix C - Minimum property schemas
18. Appendix D - Naming, identifier and namespace rules
19. Appendix E - Standards and reference-source map
20. Appendix F - Review and acceptance record

Part I - Authority, scope and conformance

1. Purpose

This artifact establishes the controlled vocabulary and semantic structure by which AIDCF describes AI capabilities and the physical, engineered, technical, operational, human and external infrastructure on which they depend. It is intended to create consistency across documents, assessments, datasets, knowledge graphs, digital twins and future tooling.

2. Objectives

- Provide a capability-first upper ontology.
- Define a vendor-neutral eight-domain taxonomy.
- Express cross-domain and external dependencies without hiding scope boundaries.
- Distinguish systems, assets, components, interfaces, resources, services, roles and evidence.
- Represent capacity, redundancy, fault domains, operating states and temporal change.
- Preserve provenance, confidence, conflict and UNKNOWN.
- Provide machine-readable semantics without requiring proprietary software.
- Support future failure analysis, control mapping, assessment and assurance decisions.

3. Scope

The ontology covers entities materially relevant to the ability of a defined AI Capability Instance to deliver specified Service Outcomes. It spans site and physical protection; facilities engineering; power and thermal systems; environmental and resource performance; building operations; accelerator compute; storage and data infrastructure; and network and interconnect architecture.

4. Exclusions

- Model quality, fairness, explainability and responsible-AI governance, except where a digital element is an infrastructure dependency.
- Application business logic, agent authority and model-to-tool trust paths, which remain primarily within AI Trust Graph.
- Legal compliance determinations and jurisdiction-specific certification decisions.
- Professional engineering approval, safety certification and life-safety design authority.
- Vendor endorsement or prescriptive selection of accelerator, fabric, storage or cooling technology.

5. Normative conformance

IOT-REQ-001 A conformant implementation shall preserve the canonical meanings and required distinctions in this artifact.

IOT-REQ-002 Extensions may add classes, relationships or properties but shall not silently redefine canonical terms.

IOT-REQ-003 Every material entity shall possess a stable identifier, class, lifecycle status, provenance and applicable confidence state.

IOT-REQ-004 Every assessment dataset shall identify the Capability Instance, Assessment Boundary, applicable Service Outcomes and Operating Envelope.

IOT-REQ-005 Dependencies crossing the Assessment Boundary shall remain represented as external dependencies or UNKNOWN; they shall not disappear because they are out of scope.

IOT-REQ-006 UNKNOWN shall remain a first-class semantic state and shall not be converted into pass, fail, zero, not applicable or accepted risk.

6. Conformance levels

Level	Name	Minimum expectation
C0	Vocabulary alignment	Uses canonical terms without contradictory meanings.
C1	Structured representation	Implements required entities, identifiers and core relationships.
C2	Assurance-ready dataset	Adds boundaries, outcomes, dependencies, states, evidence, confidence and validation.
C3	Graph-conformant implementation	Implements machine-readable schema, relationship semantics and validation shapes.
C4	Controlled extension	Publishes namespace, mappings, governance, tests and change impact for extensions.

Part II - Ontology architecture and design principles

7. Four-layer semantic architecture

Layer	Purpose	Examples
L0 Doctrine	Constrains meaning and reasoning	Capability-first; evidence-gated; cross-domain; UNKNOWN preservation.
L1 Upper ontology	Defines universal entity types	Capability, Outcome, System, Asset, Dependency, State, Actor, Evidence.
L2 Domain taxonomy	Classifies domain-specific infrastructure	UPS, CDU, accelerator node, parallel file system, scale-out fabric.
L3 Profiles and extensions	Specialises for deployment patterns	Sovereign AI, colocation, liquid-cooled training, enterprise inference.

8. Design principles

P1 Capability primacy The ontology begins with what the infrastructure must enable.

P2 Instance specificity An assessed deployment is represented as an instance, not merely a generic type.

P3 Function before product Required functions are separated from the products that implement them.

P4 Relationship explicitness Material dependencies and interfaces are encoded, not left only in narrative.

P5 State awareness Claims are tied to normal, degraded, maintenance, emergency, recovery or transition conditions.

P6 Time awareness Design, as-built, commissioned and current states are distinguishable.

P7 Evidence anchoring Material assertions can be linked to evidence and provenance.

P8 Vendor neutrality Canonical classes do not require one supplier, topology or technology.

P9 Safe extensibility Profiles can specialise the model without breaking semantic compatibility.

P10 Minimal false precision Unknown or conflicting information remains visible.

9. Metamodel

Each ontology term is described through a common metamodel.

Field	Meaning
Canonical ID	Stable AIDCF identifier.
Preferred label	Authoritative human-readable name.
Definition	Necessary and distinguishing meaning.
Entity type	Class, relationship, property, enumeration or constraint.
Parent	Immediate broader concept.
Domain	Primary AIDCF domain or cross-domain.
Allowed relations	Permitted outgoing and incoming relationships.
Required properties	Minimum data needed for conformance.
Lifecycle status	Draft, approved, deprecated or retired.
Provenance	Origin, author, approval and version.
Examples	Non-normative illustrations.
Exclusions	Concepts intentionally not covered by the term.

Part III - Upper ontology and canonical entities

10. Root entities

Class ID	Definition
AIDCF:Entity	Root of all addressable objects.
AIDCF:CapabilityEntity	An intended ability and its bounded implementation.
AIDCF:OutcomeEntity	A required or observed result.
AIDCF:InfrastructureEntity	A system, asset, component, interface, location or resource.
AIDCF:ConditionEntity	A state, constraint, assumption, limit or environmental condition.
AIDCF:AdverseEntity	A threat, hazard, initiating event, cause, failure mode or consequence.
AIDCF:AssuranceEntity	A safeguard, control, assertion, evidence, test, finding or decision.
AIDCF:PartyEntity	An organisation, actor, role, team or supplier.
AIDCF:TemporalEntity	A lifecycle stage, timestamp, validity interval or change event.

11. Canonical class definitions

Preferred term	Canonical definition
Capability	An intended operational ability supported by infrastructure.
Capability Instance	A specific deployment bounded by ownership, location, architecture, capacity, criticality and obligations.
Service Outcome	A measurable or assessable result required from a Capability Instance.
Critical Function	A function whose loss or degradation materially affects a Service Outcome.
Assessment Boundary	The declared inclusion perimeter for direct assessment.
Operating Envelope	The bounded conditions under which a capability or system is intended to operate.
Infrastructure System	A coordinated set of assets and interfaces providing one or more functions.
Asset Instance	An identifiable real-world item, service, person, procedure or resource.
Component	A constituent part of an asset or system represented where material.
Interface	A defined point or mechanism of exchange between entities.
Connection	A realised physical, logical, hydraulic, electrical, thermal or organisational link.
Resource	Consumable or constrained input required for operation.
Location	A geographic, site, building, zone, room, row, rack or logical placement.

Preferred term	Canonical definition
Dependency	A reliance by one entity upon another entity, resource, condition or decision.
Fault Domain	A set of entities expected to be affected by a common failure.
Redundancy Group	Assets or paths deliberately arranged to preserve a function.
Safeguard	A real-world measure intended to anticipate, prevent, detect, contain, respond, recover or adapt.
Evidence Object	A record or observation used to support or challenge an assertion.
Actor	A human or machine participant with a relevant role.
Organisation	A legally or operationally recognised party.
Role	A named set of responsibilities and authorities.
Operating State	A defined condition of operation.
Change Event	An approved, observed or unapproved modification affecting the represented system.

12. Mandatory distinctions

Do not collapse	Reason
Capability / Capability Instance	A generic ability is not the assessed deployment.
System / Asset / Component	Composition and assessment granularity differ.
Function / Implementing asset	A function can be reallocated or implemented by multiple technologies.
Interface / Dependency	An interface enables exchange; a dependency is reliance.
Connection / Relationship	A connection is realised; a relationship may be logical or asserted.
Capacity / Demand / Reserve	They answer different questions.
Redundancy / Diversity / Independence / Resilience	Additional paths do not prove common-cause independence or outcome resilience.
Designed / Installed / Commissioned / Observed state	Documentation and reality may diverge.
Owner / Operator / Maintainer / Risk owner	Commercial and operational responsibilities differ.
UNKNOWN / Not applicable / Ineffective	Evidence insufficiency is not the same as absence or failure.

Part IV - Capability, outcome, boundary and lifecycle model

13. Capability hierarchy

Level	Entity	Illustration
1	Capability family	AI training; inference; fine-tuning; HPC simulation; shared accelerator service.

Level	Entity	Illustration
2	Capability type	Large-scale distributed training; low-latency inference; batch inference.
3	Capability Instance	A named deployment at a defined site with a declared architecture.
4	Service Outcome	Availability, throughput, completion, latency, integrity or recovery outcome.
5	Critical Function	Heat removal, power continuity, scheduling, checkpoint persistence, fabric transport.
6	Supporting system and assets	Cooling loop, UPS path, compute cluster, storage system, network fabric.

14. Capability Instance minimum record

Property	Requirement
capabilityInstanceId	Globally or organisationally unique and immutable.
name	Human-readable identifier.
capabilityType	Controlled vocabulary value.
owner	Accountable organisation or role.
locations	Sites and zones in which it is instantiated.
lifecycleStage	Concept, design, construction, commissioning, ready, operational, change or retirement.
serviceOutcomes	One or more outcome records.
criticality	Declared and justified criticality.
assessmentBoundary	Included systems and explicit exclusions.
operatingEnvelope	Load, environmental, capacity, duration and state constraints.
architectureBaseline	Approved design or configuration reference.
effectivePeriod	Start/end or point-in-time applicability.
confidenceState	Confirmed, partial, asserted, inferred, conflicting, stale or UNKNOWN.

15. Service Outcome taxonomy

Outcome family	Examples of assessable properties
Availability	Required service continuity; tolerated interruption; degraded-mode availability.
Capacity	Minimum usable accelerator, storage, network, power and cooling capacity.
Performance	Throughput, latency, job completion, checkpoint duration, fabric efficiency.

Outcome family	Examples of assessable properties
Integrity	Correct configuration, artifact integrity, data-path integrity and state consistency.
Isolation	Tenant, cluster, management, physical and fault-domain separation.
Recoverability	Recovery objective, restart conditions, restoration sequence and validation.
Safety	Safe thermal, electrical, fire, structural and human operating conditions.
Efficiency	Resource performance within defined resilience constraints.
Sovereignty	Declared location, operational control, supply, maintenance and dependency constraints.
Observability	Minimum telemetry, alarm and diagnostic visibility needed to operate and assure.

16. Boundary model

IOT-REQ-016 An Assessment Boundary shall identify included Capability Instances, sites, zones, systems, assets, interfaces, lifecycle stages and time horizon.

IOT-REQ-017 Every exclusion shall identify rationale, owner, residual dependency and effect on confidence.

IOT-REQ-018 External services shall be represented through boundary-crossing relationships and shall not be treated as inherently reliable.

17. Lifecycle stages

Stage	Ontology emphasis
Concept	Capability intent, candidate site, constraints, assumptions and target outcomes.
Design	Required functions, systems, architecture, capacity, fault domains and interfaces.
Construction	Asset instances, installation state, deviations and provenance.
Commissioning	Tests, observations, results, outstanding defects and accepted limitations.
Operational readiness	Procedures, personnel, spares, monitoring and transfer of authority.
Operation	Current state, telemetry, maintenance, incidents, evidence and drift.
Change	Impact, configuration transition, temporary states, evidence and revalidation.
Decommissioning	Isolation, removal, sanitisation, residual hazards and record retention.

Part V - Eight-domain infrastructure taxonomy

18. Taxonomy rules

IOT-REQ-019 Domain assignment shall indicate the primary classification only; cross-domain relationships shall remain explicit.

IOT-REQ-020 Technology-specific subclasses shall inherit canonical properties from vendor-neutral parent classes.

IOT-REQ-021 A taxonomy branch shall be extended only where the distinction changes assessment, dependency, evidence, failure or assurance meaning.

19. D1 Physical Data Centre Controls

Taxonomy family	Illustrative subclasses
Site and external perimeter	Site, Campus, Property Boundary, Perimeter Barrier, Vehicle Barrier, Gate, Guard Post, External Approach, Standoff Zone
Building and internal zoning	Building, Floor, Data Hall, Room, Cage, Suite, Row, Rack, Cabinet, Security Zone, Restricted Area, Loading Area, Staging Area
Entry and identity control	Access Point, Turnstile, Mantrap, Door, Lock, Credential Reader, Biometric Reader, Visitor System, Key System
Detection and surveillance	Camera, Video Management System, Intrusion Sensor, Door Contact, Motion Sensor, Security Alarm, Intercom
Asset movement and custody	Loading Bay, Goods Lift, Screening Point, Storage Area, Seal, Chain-of-Custody Record
Physical response	Security Operations Centre, Guard Force, Response Procedure, Dispatch Channel, Incident Reconstruction Record

20. D2 Facilities Engineering

Taxonomy family	Illustrative subclasses
Civil and structural	Foundation, Floor Slab, Raised Floor, Roof, Wall, Ceiling, Rack Anchorage, Seismic Restraint, Load-Bearing Element
Electrical architecture	Incoming Supply, Substation, Transformer, Switchgear, Protection Relay, Transfer Scheme, Earthing System, Lightning Protection
Mechanical architecture	Chilled Water Plant, Condenser System, Pump System, Air Distribution, Ventilation, Pressurisation, Drainage
Fire and life safety	Detection Zone, Fire Alarm Panel, Suppression Zone, Agent System, Sprinkler System, Smoke Control, Emergency Egress, Emergency Power Off
Water and leak management	Water Source, Pipework, Valve, Drain, Sump, Leak Detection Zone, Water Treatment System
Facility control and telemetry	BMS, EPMS, SCADA, PLC, RTU, Sensor, Actuator, Gateway, Historian, Alarm Console
Engineering authority and records	Design Basis, Single-Line Diagram, P&ID, As-Built Drawing, Engineering Study, Commissioning Script, Deviation Record

21. D3 Power and Cooling Resilience

Taxonomy family	Illustrative subclasses
Utility and generation	Utility Feed, Grid Connection, On-Site Generation, Generator, Fuel Tank, Fuel System, Starting System, Exhaust System
Power conversion and distribution	Transformer, MV Switchgear, LV Switchgear, UPS, Battery, Energy Storage, Static Transfer Switch, Busway, PDU, RPP, Rack PDU, Power Supply Unit
Power quality and protection	Protection Device, Relay, Meter, Harmonic Filter, Surge Protection, Grounding, Power Quality Monitor
Heat capture	Heat Sink, Cold Plate, Rear-Door Heat Exchanger, Air Handler, In-Row Cooler, Immersion Tank
Coolant distribution	CDU, Primary Loop, Secondary Loop, Pump, Manifold, Quick Disconnect, Hose, Pipe, Heat Exchanger
Heat rejection	Chiller, Cooling Tower, Dry Cooler, Condenser, Evaporative System, Heat-Reuse Interface
Thermal and leak monitoring	Temperature Sensor, Humidity Sensor, Flow Sensor, Pressure Sensor, Leak Sensor, Coolant Quality Sensor, Thermal Camera

22. D4 Environmental Standards and Resource Performance

Taxonomy family	Illustrative subclasses
Energy resources	Grid Electricity, Renewable Electricity, On-Site Energy, Stored Energy, Fuel
Water resources	Potable Water, Process Water, Reclaimed Water, Cooling Water, Discharge Water
Environmental conditions	Ambient Temperature, Humidity, Air Quality, Particulate Level, Noise, Vibration, Flood Exposure, Heat Stress
Material and chemical resources	Refrigerant, Coolant, Fire Suppressant, Battery Chemistry, Hazardous Material
Performance indicators	PUE Record, WUE Record, CUE Record, Energy Reuse Record, Cooling Efficiency Record, Compute Utilisation Record
Environmental obligations	Permit, Reporting Requirement, Emission Limit, Water Limit, Waste Requirement, Environmental Commitment

23. D5 Building Operations

Taxonomy family	Illustrative subclasses
Governance and authority	Operating Model, Role, Delegation, Approval Authority, Escalation Path, Risk Acceptance
Work control	Procedure, Method Statement, Permit to Work, Lockout Tagout, Maintenance Window, Change Record, Temporary Configuration
Maintenance and reliability	Maintenance Plan, Work Order, Condition Monitoring, Inspection, Calibration, Spare Part, Obsolescence Record
Staffing and competence	Shift, Roster, Competency Record, Authorisation, Training Record, Contractor, Subject Matter Expert

Taxonomy family	Illustrative subclasses
Incident and emergency	Alarm, Incident, Event Log, Emergency Procedure, Crisis Team, Exercise, Post-Incident Review
Supplier and service management	Supplier, Maintenance Contract, Service Level, Remote Support, Escalation Contact, Lead Time, Supply Constraint
Configuration and documentation	Configuration Baseline, Asset Register, Drawing Register, Procedure Register, Revision, Document Custodian

24. D6 AI Compute Infrastructure

Taxonomy family	Illustrative subclasses
Accelerated compute	Accelerator, GPU, AI ASIC, Compute Node, GPU Baseboard, CPU, Memory, DPU, NIC
Rack-scale systems	Compute Tray, Rack-Scale System, Scale-Up Domain, NVLink Domain, Enclosure, Backplane, Power Shelf, Rack Controller
Cluster systems	Compute Cluster, Cluster Partition, Node Pool, Head Node, Login Node, Management Node, Provisioning Node
Management and firmware	BMC, Firmware, BIOS, Driver, Fabric Manager, Cluster Manager, Provisioning Service, Scheduler, Orchestrator
Workload infrastructure	Queue, Partition, Reservation, Container Runtime, Kubernetes Cluster, Job Scheduler, Checkpoint Agent
Compute observability	GPU Telemetry, Node Telemetry, Health Service, Event Log, Performance Counter, Error Counter
Compute lifecycle	Golden Image, Firmware Baseline, Compatibility Matrix, Hardware Support Contract, RMA Process, Secure Decommissioning Record

25. D7 AI Storage and Data Infrastructure

Taxonomy family	Illustrative subclasses
Storage systems	Parallel File System, Scale-Out NAS, Object Storage, Block Storage, Local NVMe, Burst Buffer, Archive Storage
Storage nodes and media	Storage Node, Metadata Server, Object Storage Node, Controller, SSD, HDD, Tape, NVRAM
Data-path services	Namespace, Metadata Service, Data Mover, Cache, Gateway, Client Driver, Mount Service
AI artifact services	Dataset Repository, Model Repository, Checkpoint Repository, Feature Store, Artifact Registry
Protection and recovery	Snapshot, Replica, Backup Copy, Erasure Coding Group, Recovery Copy, Immutable Copy
Storage management	Storage Manager, Capacity Manager, Tiering Policy, Retention Policy, Integrity Scanner, Telemetry Service
Data logistics interfaces	Ingestion Interface, Export Interface, Data Transfer Node, WAN Transfer Service, Staging Area

26. D8 AI Network and Interconnect Architecture

Taxonomy family	Illustrative subclasses
Scale-up interconnect	Scale-Up Fabric, Switch Fabric, Interconnect Link, Topology Domain
Scale-out compute fabric	InfiniBand Fabric, Ethernet AI Fabric, RoCE Fabric, Compute Leaf, Compute Spine, Fabric Manager
Storage fabric	Storage Network, Storage Leaf, Storage Spine, SAN Fabric, Storage Gateway
Frontend and service network	Frontend Network, Load Balancer, Service Gateway, Edge Router, Firewall, API Network
Management and control network	Management Network, Out-of-Band Network, Console Server, Management Switch, Jump Service
Timing and core services	Time Service, DNS Service, DHCP Service, IPAM Service, Certificate Service, Authentication Service
Physical connectivity	Cable, Optical Fibre, Copper Link, Transceiver, Patch Panel, Meet-Me Room, Cross-Connect
Network observability	Flow Telemetry, Fabric Telemetry, Packet Capture Service, Network Controller, Congestion Monitor, Path Monitor

Part VI - Cross-cutting classification facets

27. Facet model

Domain classification alone is insufficient. Every material entity may also be classified through orthogonal facets that support analysis without creating duplicate classes.

Facet	Controlled values / meaning
Criticality	Critical, High, Moderate, Low, Unclassified, UNKNOWN.
Lifecycle status	Planned, designed, procured, installed, commissioned, operational, standby, degraded, under maintenance, bypassed, failed, recovering, retired.
Physicality	Physical, logical, human, procedural, contractual, environmental, informational.
Boundary position	Internal, boundary-crossing, external, excluded-but-dependent, UNKNOWN.
Tenancy	Dedicated, shared, multi-tenant, pooled, reserved, burst, UNKNOWN.
Control authority	Direct, delegated, contractual, advisory, none, UNKNOWN.
Evidence status	Confirmed, partially confirmed, asserted, inferred, conflicting, stale, UNKNOWN.
Replaceability	Immediate, stocked spare, contractually replaceable, long-lead, irreplaceable, UNKNOWN.
Mobility	Fixed, relocatable, portable, virtualised, service-based.
Exposure	Public, campus, building, hall, cage, rack, management plane, isolated, UNKNOWN.
Sovereignty relevance	Local control, foreign dependency, cross-border support, remote administration, jurisdiction-sensitive, not declared.
Environmental sensitivity	Temperature, humidity, water quality, vibration, particulate, altitude, electromagnetic, acoustic.

28. Criticality semantics

IOT-REQ-028 Criticality shall be assigned in relation to a Capability Instance and Service Outcome, not as an intrinsic universal property of an asset class.

IOT-REQ-029 Criticality rationale should consider consequence, time sensitivity, substitutability, concentration, recovery complexity, safety, regulatory and strategic significance.

29. Classification inheritance

A subclass inherits properties and permitted relationships from its parent unless an approved extension explicitly narrows or adds semantics. Inheritance shall not be used to infer an operational fact about an instance. For example, classifying an asset as a UPS does not establish its capacity, redundancy, condition or effectiveness.

Part VII - Asset, system, interface and resource model

30. System decomposition

Construct	Use
Infrastructure System	Group assets that collectively provide a function.
Subsystem	Represent a meaningful functional subdivision.
Asset Group	Group similar instances for inventory or analysis.
Functional Group	Group heterogeneous assets that realise a function.
Redundancy Group	Represent alternatives designed to preserve a function.
Fault Domain	Represent common failure exposure.
Protection Zone	Represent a security, fire, electrical or thermal boundary.
Maintenance Group	Represent assets taken out of service together.
Capacity Pool	Represent shared capacity available to consumers.

31. Interface taxonomy

Interface class	Examples	Minimum properties
Electrical	Supply, feeder, bus, breaker termination	Voltage, phase, capacity, protection, source, destination.
Thermal	Air or liquid heat transfer point	Medium, temperature, flow, pressure, capacity, direction.
Hydraulic	Coolant or water connection	Fluid, pressure, flow, valve/isolation, leak boundary.
Network	Port, link, VLAN, subnet, fabric endpoint	Protocol, bandwidth, topology role, path, security zone.
Mechanical	Mounting, anchoring, ventilation or structural contact	Load, tolerance, connection, isolation.
Control	Sensor, command, alarm or automation interface	Signal, authority, fail state, protocol, authentication.
Human/procedural	Handover, approval, escalation or work interface	Role, trigger, response objective, record.
Contractual/service	Utility, supplier or maintenance service boundary	Provider, service level, capacity, escalation, evidence.

32. Resource taxonomy

Resource	Key semantics
Electrical energy	Source, quality, capacity, duration, carbon attribute, dependency.
Fuel	Type, quantity, quality, replenishment, usable duration.
Cooling capacity	Heat-removal capacity under defined conditions.

Resource	Key semantics
Water	Source, quality, quantity, permitted use and continuity.
Coolant/refrigerant	Type, compatibility, quantity, quality and handling constraints.
Space and structural capacity	Area, volume, floor loading, rack loading and access.
Network bandwidth	Usable throughput, latency, oversubscription and state.
Storage capacity	Usable capacity, performance tier, protection overhead and headroom.
Compute capacity	Usable accelerator/CPU/memory capacity and allocation.
Human capacity	Competence, availability, authorisation and concentration.
Time	Transfer time, autonomy, response time, maintenance window, recovery time.
Spares	Quantity, compatibility, location, lead time and preservation state.

33. Minimum asset record

Property	Cardinality	Meaning
assetId	Required	Stable identifier.
assetClass	Required	Canonical or approved extension class.
name	Required	Human-readable name.
systemMembership	Required where applicable	Parent system(s).
location	Required for physical assets	Placement or service jurisdiction.
owner	Required	Accountable owner or UNKNOWN.
operator	Required for operational assets	Operator or UNKNOWN.
maintainer	Conditional	Maintenance party.
lifecycleStatus	Required	Controlled value.
criticality	Required if material	Capability-specific value and rationale.
capacityProperties	Conditional	Units and operating conditions.
interfaces	Conditional	Connections and service boundaries.
dependencies	Required if material	Direct and known indirect reliances.
faultDomains	Required if resilience-relevant	One or more fault-domain memberships.
evidenceLinks	Required for assurance use	Evidence supporting material attributes.
validity	Required	Effective period and observation date.
confidenceState	Required	Confirmed through UNKNOWN.

Part VIII - Relationship and dependency vocabulary

34. Relationship design rules

IOT-REQ-034 Every relationship shall identify subject, predicate and object; material relationships shall also identify scope, conditions, provenance, confidence and effective period.

IOT-REQ-035 Inverse relationships may be generated only where the canonical registry declares an inverse.

IOT-REQ-036 Symmetric relationships shall not be assumed for directional dependencies.

IOT-REQ-037 Transitive reasoning shall be applied only to approved predicates and shall not automatically infer engineering sufficiency.

35. Core structural relationships

Predicate	Subject	Object	Logic	Inverse	Meaning
PART_OF	Entity	Entity	Transitive	HAS_PART	Composition.
CONTAINS	Location/System	Entity	Transitive	LOCATED_IN/PART_OF	Containment.
LOCATED_IN	Physical Entity	Location	Transitive	CONTAINS	Physical placement.
MEMBER_OF	Entity	Group	No	HAS_MEMBER	Group membership.
IMPLEMENTS	Asset/System	Function	No	IMPLEMENTED_BY	Realises a required function.
SUPPORTS	Entity	Capability/Outcome	No	SUPPORTED_BY	Contributes to outcome.
INTERFACES_WITH	Entity	Entity	Symmetric	INTERFACES_WITH	Defined exchange boundary.
CONNECTED_TO	Entity	Entity	Symmetric unless directed	CONNECTED_TO	Realised connection.
ROUTES_THROUGH	Flow/Service	Entity	No	CARRIES	Path relationship.

36. Functional dependency relationships

Predicate	Subject	Object	Meaning
DEPENDS_ON	Consumer entity	Provider/entity/condition	General reliance.
POWERED_BY	Load/system	Electrical source/path	Electrical reliance.
COOLED_BY	Heat source/system	Cooling system/path	Thermal reliance.
SUPPLIED_BY	Consumer	Provider/resource	Resource or service supply.
CONTROLLED_BY	Asset/system	Controller/actor	Operational or automated control.
MONITORED_BY	Entity	Monitoring entity	Visibility and detection reliance.
PROTECTED_BY	Entity	Safeguard	Protective relationship.
AUTHENTICATED_BY	Entity/service	Identity service	Identity reliance.
TIMED_BY	Entity/service	Time source	Timing reliance.
CONFIGURED_BY	Entity	Configuration source/actor	Configuration reliance.

Predicate	Subject	Object	Meaning
RECOVERED_BY	Entity/capability	Recovery asset/process	Restoration reliance.
MAINTAINED_BY	Entity	Actor/organisation	Maintenance reliance.
OPERATED_BY	Entity/system	Actor/organisation	Operational responsibility.
EVIDENCED_BY	Assertion/entity	Evidence object	Evidence support.
TESTED_BY	Assertion/safeguard	Test record	Validation link.

37. Dependency record schema

Field	Required	Description
dependencyId	Yes	Stable relationship identifier.
consumer	Yes	Entity reliant on another.
provider	Yes or UNKNOWN	Entity, resource or condition relied upon.
dependencyType	Yes	Electrical, thermal, hydraulic, network, control, human, supplier, etc.
strength	Yes	Absolute, material, conditional, supporting, recovery-only or UNKNOWN.
direction	Yes	Provider to consumer effect direction.
applicableStates	Yes	Operating states in which reliance is material.
capacityRequirement	Conditional	Required quantity, unit, duration and quality.
substitutability	Conditional	Alternative provider or workaround.
transferMechanism	Conditional	Automatic, manual, procedural or none.
commonDependencies	Conditional	Shared providers or causes.
owner	Yes or UNKNOWN	Responsible party for the dependency.
evidence	Yes for assurance	Supporting documents, observations or tests.
confidence	Yes	Confidence state and rationale.
validity	Yes	Effective period and validation date.

38. Dependency classes

Class	Definition	Examples
Electrical	Reliance on electrical supply, quality, protection or distribution.	GPU rack → rack PDU → busway → UPS path.
Thermal	Reliance on heat capture, transport or rejection.	GPU tray → cold plate → secondary loop → CDU.
Hydraulic	Reliance on fluid path, pressure, flow or quality.	CDU → facility water loop.
Mechanical/structural	Reliance on physical support, airflow or mechanical function.	Rack-scale system → floor loading and anchoring.
Network	Reliance on connectivity, bandwidth, route or fabric service.	Compute node → scale-out fabric.

Class	Definition	Examples
Control-system	Reliance on sensors, automation, logic or command.	Cooling valve → PLC → sensor chain.
Timing/information	Reliance on time, name, address, telemetry or configuration.	Cluster service → DNS/NTP/IPAM.
Human/procedural	Reliance on competent action, approval, handover or procedure.	Generator response → operator and escalation.
Supplier/contractual	Reliance on external service, warranty, spares or service level.	Storage recovery → OEM support.
Geographic/environmental	Reliance on location, ambient or external condition.	Heat rejection → outdoor wet-bulb condition.
Capacity	Reliance on sufficient headroom or reserve.	Cluster expansion → resilient power headroom.
Recovery-only	Reliance primarily during restoration.	Compute rebuild → golden image repository.

39. Common-mode and concentration relationships

SHARES_DEPENDENCY_WITH Two entities rely on the same material provider.

IN_SAME_FAULT_DOMAIN_AS Two entities are exposed to a declared common fault domain.

DIVERSE_FROM Two implementations differ in a declared material characteristic.

INDEPENDENT_OF An evidence-backed claim of relevant independence under stated conditions.

REDUNDANT_WITH An alternative intended to preserve a function.

CONCENTRATES An entity aggregates capability, capacity or dependency consequence.

SINGLE_POINT_FOR Failure of an entity defeats a required outcome without effective alternative.

ISOLATED_FROM A tested or designed separation relevant to a declared hazard or threat.

Part IX - Capacity, redundancy, fault-domain and resilience semantics

40. Capacity model

Capacity term	Canonical meaning
Nameplate capacity	Manufacturer or design rating before contextual adjustment.
Installed capacity	Capacity physically present.
Commissioned capacity	Capacity demonstrated within commissioning scope.
Available capacity	Capacity currently usable after condition, derating and outage.
Committed capacity	Capacity allocated, contracted or reserved.
Demand	Observed or forecast requirement under stated condition.
Headroom	Available capacity less demand and commitments, subject to constraints.

Capacity term	Canonical meaning
Resilient capacity	Capacity available during a defined adverse or maintenance condition.
Stranded capacity	Installed capacity unusable because another dependency constrains it.
Transient capacity	Capacity supportable for a limited duration.
Recovery capacity	Capacity available while restoration is underway.
UNKNOWN capacity	Capacity cannot be established with sufficient evidence.

41. Capacity assertion schema

IOT-REQ-041 A capacity value shall identify metric, unit, entity, operating state, environmental conditions, duration, derating assumptions, evidence date and confidence.

IOT-REQ-042 Nameplate, installed, commissioned, available, committed, resilient and uncommitted capacity shall not be used interchangeably.

IOT-REQ-043 A cross-domain capacity claim shall identify the limiting dependency and any stranded capacity.

42. Redundancy semantics

Concept	Required elements
Redundancy claim	Protected function, alternatives, required capacity, failure condition, transfer mechanism, transfer time and evidence.
Diversity claim	Characteristic in which alternatives materially differ, such as route, technology, supplier or control.
Independence claim	Relevant common causes considered, shared dependencies identified, applicable states and evidence.
Maintainability claim	Function sustained while defined assets are removed from service safely.
Concurrent-maintenance claim	Specific maintenance scenario, capacity, operational restrictions and evidence.
Fault tolerance claim	Failure set, containment boundary, residual outcome and evidence.
Graceful degradation	Priority, reduced outcome, trigger, control mechanism and recovery path.

43. Fault-domain taxonomy

- Physical: room, rack, route, fire zone, flood zone or structural area.
- Electrical: utility source, transformer, switchboard, UPS path, busway or protection zone.
- Thermal: air zone, coolant loop, CDU, chiller train, heat-rejection path.
- Network: switch, plane, fabric, route, controller or failure group.
- Storage: node group, enclosure, metadata service, protection set or site.
- Compute: node, tray, rack, scale-up domain, cluster partition or scheduler domain.
- Control-system: sensor chain, controller, automation logic, communication path or management platform.
- Software/configuration: firmware baseline, driver, image, policy or configuration source.
- Human/operational: team, shift, procedure, supplier or authority concentration.
- External: utility, carrier, water source, fuel supply, vendor or geographic region.

44. Resilience claim record

Property	Meaning
subject	Capability, outcome, function or system.
adverseCondition	Failure, threat, hazard, maintenance or compound scenario.
initialState	State before the event.
requiredOutcome	Minimum acceptable service result.
permittedDegradation	Allowed reduction and duration.
safeguards	Measures expected to act.
responseSequence	Detection, decision, transfer, containment and recovery sequence.
recoveryObjective	Target restoration condition.
evidence	Tests, events, analysis or observations.
limitations	Uncovered states, assumptions and constraints.
confidence	Supported conclusion and rationale.

Part X - Operating state, temporal state and change semantics

45. Operating state taxonomy

Code	Definition
NORMAL	Approved architecture operates within declared envelope; no material planned impairment.
DEGRADED	One or more functions, safeguards, paths or reserves are impaired while some capability remains.
MAINTENANCE	A planned intervention changes availability, configuration, staffing or safeguards.
EMERGENCY	Exceptional condition requiring urgent protective action.
RECOVERY	Controlled restoration and validation after disruption.
TRANSITION	Movement between architectures, states, locations or control regimes.
SHUTDOWN	Intentional cessation with controlled residual dependencies.
FAILED	Required function is not delivered.
UNKNOWN	Current operating state cannot be determined.

46. State transition model

IOT-REQ-046 State transitions shall identify trigger, authority, preconditions, required safeguards, expected duration, monitoring, exit criteria and evidence.

IOT-REQ-047 Temporary bypass and maintenance configurations shall be represented as explicit states or configuration versions.

IOT-REQ-048 Assurance claims shall identify the operating states they cover and shall not be generalised to uncovered states.

47. Temporal truth

Temporal construct	Use
Observation time	When a fact was observed.
Effective time	When a fact or relationship applies in the represented system.
Record time	When the information entered the repository.
Validity interval	Start and end of applicability.
Evidence period	Period covered by evidence.
Review date	When a record was last assessed.
Expiry date	When evidence or an assertion is no longer presumed current.

Temporal construct	Use
Supersedes	Links a new record or baseline to the prior one.
Derived from	Links a conclusion to source records.

48. Configuration perspectives

Perspective	Meaning
Intended	Approved requirement or design intent.
Designed	Documented architecture.
Procured	Contracted or ordered implementation.
Installed	Physical or logical implementation present.
As-built	Recorded installed configuration.
Commissioned	Configuration subjected to defined acceptance tests.
Observed	State directly observed or measured.
Current approved	Authoritative operational baseline.
Temporary	Time-bounded deviation or workaround.
Future	Planned target state.
Retired	No longer active but retained for history and dependency closure.

49. Change Event schema

Field	Description
changeId	Stable identifier.
changeType	Addition, removal, replacement, upgrade, reconfiguration, relocation, capacity change, supplier change or emergency change.
affectedEntities	Assets, systems, interfaces, outcomes and dependencies.
preState/postState	Configuration and operating state before and after.
impactAnalysis	Affected capacity, fault domains, safeguards, evidence and claims.
authority	Approver and implementer.
plannedWindow	Applicable schedule if planned.
validation	Required testing and acceptance.
rollback	Restoration path and conditions.
assuranceImpact	Claims invalidated, limited or requiring reassessment.

Part XI - Responsibility, evidence, confidence and UNKNOWN semantics

50. Responsibility model

Relationship	Meaning
OWNS	Legal, financial or accountable ownership.
OPERATES	Performs routine operation.
CONTROLS	Holds authority to command or configure.
MAINTAINS	Performs inspection, servicing and repair.
SUPPLIES	Provides a resource, product or service.
MONITORS	Observes state and raises information or alerts.
APPROVES	Authorises design, change, work or decision.
ASSURES	Performs independent or internal assurance activity.
ACCEPTS_RISK_FOR	Holds authority to accept residual exposure.
CUSTODIAN_OF	Maintains an evidence or information object.
RESPONDS_TO	Has response responsibility for an event or alarm.
ESCALATES_TO	Transfers information or decision authority.

51. Shared-responsibility record

IOT-REQ-051 A shared-responsibility allocation shall identify function, assets, lifecycle stages, operating states, responsible party, accountable party, consulted parties, evidence custodian and unresolved gaps.

IOT-REQ-052 Contractual responsibility shall not be treated as proof of operational capability or control effectiveness.

52. Evidence attachment model

Evidence relation	Meaning
SUPPORTS_ASSERTION	Evidence supports a testable statement.
CHALLENGES_ASSERTION	Evidence contradicts or weakens a statement.
OBSERVES_ENTITY	Evidence records the state of an entity.
VALIDATES_SAFEGUARD	Evidence demonstrates a safeguard under stated conditions.
ESTABLISHES_PROVENANCE	Evidence identifies source, origin or custody.
COVERS_PERIOD	Evidence applies to a declared period.
COVERS_STATE	Evidence applies to one or more operating states.
CONFLICTS_WITH	Evidence is inconsistent with another evidence object.

Evidence relation	Meaning
SUPERSEDES	Newer controlled evidence replaces an earlier record.

53. Confidence states

State	Meaning
CONFIRMED	Direct, relevant and sufficiently current evidence supports the assertion.
PARTIALLY_CONFIRMED	Evidence supports part of the assertion or scope.
ASSERTED	Claim provided by an accountable source but not independently corroborated.
INFERRED	Derived from related facts using declared reasoning.
CONFLICTING	Material sources disagree and conflict remains unresolved.
STALE	Evidence or observation is no longer sufficiently current.
UNKNOWN	Information is insufficient to reach a defensible state.

54. UNKNOWN doctrine

IOT-REQ-054 UNKNOWN may apply to an entity existence, class, property, relationship, state, capacity, owner, evidence status or conclusion.

IOT-REQ-055 UNKNOWN shall identify the missing information, why it matters, current owner for resolution where known, and assurance impact.

IOT-REQ-056 UNKNOWN shall not be replaced with an assumed favourable value to enable scoring or certification.

55. Conflict representation

Conflicting records shall be retained with separate provenance and effective periods. A resolution record may select, reconcile or reject assertions, but the original conflict trail should remain auditable.

Part XII - Conformance rules and validation shapes

56. Core validation rules

Rule	Validation statement
VAL-001	Every Capability Instance has at least one Service Outcome.
VAL-002	Every Service Outcome is supported by at least one Critical Function.
VAL-003	Every Critical Function is implemented by at least one System, Asset or external service, or is UNKNOWN.
VAL-004	Every material physical Asset has a Location or UNKNOWN.
VAL-005	Every material Asset has an owner and operator or explicit UNKNOWN.
VAL-006	Every boundary-crossing dependency identifies an external provider or UNKNOWN.
VAL-007	Every capacity assertion has metric, unit, state, condition, time and confidence.
VAL-008	Every redundancy claim identifies protected function, alternatives, transfer mechanism, common dependencies and evidence.
VAL-009	Every independence claim is evidence-backed and condition-bounded.
VAL-010	Every critical dependency identifies applicable states and strength.
VAL-011	Every assurance-relevant relationship has provenance and effective period.
VAL-012	Every asset class derives from a canonical or registered extension class.
VAL-013	Every Change Event links affected entities and assurance impact.
VAL-014	Every UNKNOWN record states the unresolved attribute and its consequence.
VAL-015	Every Service Outcome and resilience claim is tied to a Capability Instance and Operating Envelope.
VAL-016	No evidence object is treated as a safeguard; it evidences a safeguard or assertion.
VAL-017	No control abstraction is represented as proof that a real safeguard exists.
VAL-018	No maturity value automatically creates an assurance decision.
VAL-019	No excluded dependency is removed from the capability dependency view.
VAL-020	No vendor-specific term replaces its canonical parent in cross-profile exchange.

57. Quality tests

Test family	Questions
Completeness	Are material capabilities, outcomes, functions, assets, external services and dependencies represented?
Consistency	Do class and relationship uses obey definitions and cardinalities?
Traceability	Can each material assertion be traced to evidence and provenance?
Currency	Do effective dates and configuration states reflect the assessment point in time?
Boundary integrity	Are exclusions and boundary-crossing dependencies visible?
Cross-domain integration	Do power, cooling, control, compute, storage and network chains connect?
Failure readiness	Can Artifact 4 attach causes, failure modes, propagation and consequences without semantic gaps?
Assurance readiness	Can safeguards, evidence, findings and decisions attach to the represented subject?
Interoperability	Can the dataset be exported using controlled identifiers and namespaces?
Human usability	Can engineers and assessors interpret the model without specialist graph tooling?

58. Prohibited modelling practices

- Using a single building-level rating as a property of every hosted capability.
- Treating product type as proof of performance or compliance.
- Hiding external dependencies because they are contractually assigned.
- Representing nominal redundancy without shared-dependency analysis.
- Combining designed, as-built and observed states into one unqualified record.
- Encoding UNKNOWN as zero, false, pass, fail or not applicable.
- Creating vendor-specific top-level classes where a vendor-neutral parent is sufficient.
- Using transitive graph reasoning to infer capacity, safety or engineering adequacy.
- Removing superseded records needed to understand historical decisions.
- Treating an ontology completeness score as infrastructure assurance.

Part XIII - Reference implementation profiles

59. Profile construction method

A profile selects mandatory classes, relationships, properties, evidence expectations and validation rules for a repeatable deployment pattern. Profiles shall not alter upper ontology meaning.

Profile	Scope and emphasis
Enterprise inference facility	Low- or moderate-scale inference in an enterprise site. Emphasis: latency, service networks, shared power/cooling, tenant interaction, recoverability.
Distributed training cluster	Multi-node accelerated training. Emphasis: scale-up/scale-out fabrics, checkpoint storage, job completion, power/thermal concentration and failure containment.

Profile	Scope and emphasis
Liquid-cooled rack-scale AI system	High-density rack-scale system. Emphasis: direct-liquid loops, CDU dependencies, water quality, leak boundaries, transient response, rack power and structural load.
AI colocation tenancy	Tenant equipment hosted in provider facility. Emphasis: boundary, responsibility, evidence access, contracted capacity, shared fault domains and cross-connects.
Sovereign AI infrastructure	Strategic capability with location and control constraints. Emphasis: operational autonomy, remote support, jurisdiction, supply, spares, identities and external services.
AI supercomputing facility	Large HPC/AI deployment. Emphasis: high-performance fabric, parallel storage, scheduler, job resilience, large-scale power/cooling and recovery.
Edge AI facility	Constrained remote deployment. Emphasis: environmental envelope, limited staffing, network dependence, remote management and local safety.
Multi-site AI service	Capability distributed across sites. Emphasis: inter-site dependencies, data replication, routing, control plane, failover and service-level boundary.

60. Worked profile: liquid-cooled distributed training capability

Element	Illustrative ontology record
Capability Instance	Distributed foundation-model training capability at Site A.
Service Outcomes	Minimum usable accelerator capacity; maximum interruption; checkpoint survival; bounded restart time.
Critical Functions	Electrical continuity; heat capture and rejection; scale-out transport; metadata access; checkpoint persistence; scheduling.
Systems	Compute cluster; rack power path; liquid cooling system; parallel storage; AI fabric; management system.
Boundary	Defined halls, racks, infrastructure systems and external utility/OEM services.
Cross-domain dependencies	Utility → UPS → busway → rack; cold plate → secondary loop → CDU → facility loop → heat rejection; node → fabric → storage.
Fault domains	Rack-scale compute domain; CDU loop; electrical path; fabric plane; storage protection group; operator team.
Operating states	Normal, single-path maintenance, degraded cooling, recovery and transition.
Critical UNKNOWN examples	Unverified secondary-loop autonomy; unclear shared controller; stale resilient-capacity study.
Evidence links	As-built diagrams, commissioning tests, telemetry, maintenance records, failure tests and observed configuration.

61. Competency questions for profile review

- Can every required service outcome be traced to functions, systems and assets?
- Are physical and logical topology views consistent?

- Are power and cooling capacities expressed under the same operating conditions as compute demand?
- Are management and control dependencies represented, including those required during recovery?
- Are shared dependencies visible across apparent redundant paths?
- Does evidence cover degraded and maintenance states, not only normal operation?
- Can an external reviewer identify what remains UNKNOWN without reverse-engineering narrative reports?

Part XIV - Machine-readable publication specification

62. Required publication package

File	Purpose
aidcf-iot-v1.0.owl	Formal ontology in OWL.
aidcf-iot-v1.0.ttl	Human-readable RDF/Turtle serialization.
aidcf-context-v1.0.jsonld	JSON-LD context.
aidcf-schema-v1.0.json	JSON Schema for exchange records.
aidcf-shapes-v1.0.ttl	SHACL validation shapes.
aidcf-vocabulary-v1.0.csv	Controlled term register.
aidcf-relationships-v1.0.csv	Relationship registry.
aidcf-example-minimal-v1.0.jsonld	Minimal conformant example.
aidcf-example-training-v1.0.jsonld	Worked training-cluster profile.
CHANGELOG.md	Version history and changes.
MIGRATION.md	Compatibility and migration guidance.
LICENSE.md	Usage and extension rules.

63. Namespace policy

Recommended public namespace pattern: https://w3id.org/aidcf/iot/1.0/. A permanent redirect service or another governed persistent identifier may be selected before publication. Draft namespaces shall be clearly marked and shall not be represented as stable production identifiers.

64. Identifier rules

- Class identifiers use PascalCase, for example CapabilityInstance.
- Relationship identifiers use UPPER_SNAKE_CASE in human registries and stable lowerCamelCase or IRI form in machine schemas.
- Instance identifiers shall be opaque enough to avoid embedding sensitive facility details.
- Identifiers shall not be reassigned after retirement.
- Human-readable labels may change without changing semantic identifiers where meaning is stable.
- Breaking semantic changes require a new major namespace version.

65. JSON exchange skeleton

```
{
  "@context": "https://w3id.org/aidcf/iot/1.0/context.jsonld",
  "id": "capability-instance:example-001",
  "type": "CapabilityInstance",
  "name": "Example distributed training capability",
  "serviceOutcomes": ["outcome:compute-capacity", "outcome:checkpoint-recovery"],
  "assessmentBoundary": "boundary:example-001",
  "operatingEnvelope": "envelope:example-001",
  "supportedBy": ["system:compute-cluster", "system:cooling-loop", "system:power-path"],
  "confidenceState": "ASSERTED"
}
```

66. Extension governance

IOT-REQ-066 An extension shall publish namespace, owner, scope, parent mappings, definitions, constraints, examples, tests, version and compatibility statement.

IOT-REQ-067 A profile shall not weaken canonical validation rules without declaring a nonconformance.

IOT-REQ-068 Vendor mappings should map product terms to canonical classes rather than creating competing upper ontologies.

67. Integration boundary with AI Trust Graph

AIDCF primary semantics	Shared handoff	AI Trust Graph primary semantics
Site, building, power, cooling, facility OT, physical assets, compute hardware, storage infrastructure and interconnect.	Identifiers, evidence, infrastructure services, machine identities, management interfaces, dependencies and findings.	Models, agents, tools, APIs, human and machine identities, authority, reachability and digital trust paths.

Appendix A - Canonical class registry

Canonical ID	Preferred label	Definition / use	Domain
AIDCF:Capability	Capability	An intended operational ability supported by infrastructure.	Core
AIDCF:CapabilityInstance	Capability Instance	A specific deployment bounded by ownership, location, architecture, capacity, criticality and obligations.	Core
AIDCF:ServiceOutcome	Service Outcome	A measurable or assessable result required from a Capability Instance.	Core
AIDCF:CriticalFunction	Critical Function	A function whose loss or degradation materially affects a Service Outcome.	Core
AIDCF:AssessmentBoundary	Assessment Boundary	The declared inclusion perimeter for direct assessment.	Core
AIDCF:OperatingEnvelope	Operating Envelope	The bounded conditions under which a capability or system is intended to operate.	Core
AIDCF:InfrastructureSystem	Infrastructure System	A coordinated set of assets and interfaces providing one or more functions.	Core
AIDCF:AssetInstance	Asset Instance	An identifiable real-world item, service, person, procedure or resource.	Core
AIDCF:Component	Component	A constituent part of an asset or system represented where material.	Core
AIDCF:Interface	Interface	A defined point or mechanism of exchange between entities.	Core
AIDCF:Connection	Connection	A realised physical, logical, hydraulic, electrical, thermal or organisational link.	Core
AIDCF:Resource	Resource	Consumable or constrained input required for operation.	Core
AIDCF:Location	Location	A geographic, site, building, zone, room, row, rack or logical placement.	Core
AIDCF:Dependency	Dependency	A reliance by one entity upon another entity, resource, condition or decision.	Core
AIDCF:FaultDomain	Fault Domain	A set of entities expected to be affected by a common failure.	Core
AIDCF:RedundancyGroup	Redundancy Group	Assets or paths deliberately arranged to preserve a function.	Core
AIDCF:Safeguard	Safeguard	A real-world measure intended to anticipate, prevent, detect, contain, respond, recover or adapt.	Core
AIDCF:EvidenceObject	Evidence Object	A record or observation used to support or challenge an assertion.	Core
AIDCF:Actor	Actor	A human or machine participant with a relevant role.	Core
AIDCF:Organisation	Organisation	A legally or operationally recognised party.	Core
AIDCF:Role	Role	A named set of responsibilities and authorities.	Core
AIDCF:OperatingState	Operating State	A defined condition of operation.	Core

Canonical ID	Preferred label	Definition / use	Domain
AIDCF:ChangeEvent	Change Event	An approved, observed or unapproved modification affecting the represented system.	Core
AIDCF:Site	Site	Site as classified within Site and external perimeter.	D1
AIDCF:Campus	Campus	Campus as classified within Site and external perimeter.	D1
AIDCF:PropertyBoundary	Property Boundary	Property Boundary as classified within Site and external perimeter.	D1
AIDCF:PerimeterBarrier	Perimeter Barrier	Perimeter Barrier as classified within Site and external perimeter.	D1
AIDCF:VehicleBarrier	Vehicle Barrier	Vehicle Barrier as classified within Site and external perimeter.	D1
AIDCF:Gate	Gate	Gate as classified within Site and external perimeter.	D1
AIDCF:GuardPost	Guard Post	Guard Post as classified within Site and external perimeter.	D1
AIDCF:ExternalApproach	External Approach	External Approach as classified within Site and external perimeter.	D1
AIDCF:StandoffZone	Standoff Zone	Standoff Zone as classified within Site and external perimeter.	D1
AIDCF:Building	Building	Building as classified within Building and internal zoning.	D1
AIDCF:Floor	Floor	Floor as classified within Building and internal zoning.	D1
AIDCF:DataHall	Data Hall	Data Hall as classified within Building and internal zoning.	D1
AIDCF:Room	Room	Room as classified within Building and internal zoning.	D1
AIDCF:Cage	Cage	Cage as classified within Building and internal zoning.	D1
AIDCF:Suite	Suite	Suite as classified within Building and internal zoning.	D1
AIDCF:Row	Row	Row as classified within Building and internal zoning.	D1
AIDCF:Rack	Rack	Rack as classified within Building and internal zoning.	D1
AIDCF:Cabinet	Cabinet	Cabinet as classified within Building and internal zoning.	D1
AIDCF:SecurityZone	Security Zone	Security Zone as classified within Building and internal zoning.	D1
AIDCF:RestrictedArea	Restricted Area	Restricted Area as classified within Building and internal zoning.	D1
AIDCF:LoadingArea	Loading Area	Loading Area as classified within Building and internal zoning.	D1
AIDCF:StagingArea	Staging Area	Staging Area as classified within Building and internal zoning.	D1
AIDCF:AccessPoint	Access Point	Access Point as classified within Entry and identity control.	D1
AIDCF:Turnstile	Turnstile	Turnstile as classified within Entry and identity control.	D1

Canonical ID	Preferred label	Definition / use	Domain
AIDCF:Mantrap	Mantrap	Mantrap as classified within Entry and identity control.	D1
AIDCF:Door	Door	Door as classified within Entry and identity control.	D1
AIDCF:Lock	Lock	Lock as classified within Entry and identity control.	D1
AIDCF:CredentialReader	Credential Reader	Credential Reader as classified within Entry and identity control.	D1
AIDCF:BiometricReader	Biometric Reader	Biometric Reader as classified within Entry and identity control.	D1
AIDCF:VisitorSystem	Visitor System	Visitor System as classified within Entry and identity control.	D1
AIDCF:KeySystem	Key System	Key System as classified within Entry and identity control.	D1
AIDCF:Camera	Camera	Camera as classified within Detection and surveillance.	D1
AIDCF:VideoManagementSystem	Video Management System	Video Management System as classified within Detection and surveillance.	D1
AIDCF:IntrusionSensor	Intrusion Sensor	Intrusion Sensor as classified within Detection and surveillance.	D1
AIDCF:DoorContact	Door Contact	Door Contact as classified within Detection and surveillance.	D1
AIDCF:MotionSensor	Motion Sensor	Motion Sensor as classified within Detection and surveillance.	D1
AIDCF:SecurityAlarm	Security Alarm	Security Alarm as classified within Detection and surveillance.	D1
AIDCF:Intercom	Intercom	Intercom as classified within Detection and surveillance.	D1
AIDCF:LoadingBay	Loading Bay	Loading Bay as classified within Asset movement and custody.	D1
AIDCF:GoodsLift	Goods Lift	Goods Lift as classified within Asset movement and custody.	D1
AIDCF:ScreeningPoint	Screening Point	Screening Point as classified within Asset movement and custody.	D1
AIDCF:StorageArea	Storage Area	Storage Area as classified within Asset movement and custody.	D1
AIDCF:Seal	Seal	Seal as classified within Asset movement and custody.	D1
AIDCF:Chain-of-CustodyRecord	Chain-of-Custody Record	Chain-of-Custody Record as classified within Asset movement and custody.	D1
AIDCF:SecurityOperationsCentre	Security Operations Centre	Security Operations Centre as classified within Physical response.	D1
AIDCF:GuardForce	Guard Force	Guard Force as classified within Physical response.	D1
AIDCF:ResponseProcedure	Response Procedure	Response Procedure as classified within Physical response.	D1
AIDCF:DispatchChannel	Dispatch Channel	Dispatch Channel as classified within Physical response.	D1
AIDCF:IncidentReconstructionRecord	Incident Reconstruction Record	Incident Reconstruction Record as classified within Physical response.	D1

Canonical ID	Preferred label	Definition / use	Domain
AIDCF:Foundation	Foundation	Foundation as classified within Civil and structural.	D2
AIDCF:FloorSlab	Floor Slab	Floor Slab as classified within Civil and structural.	D2
AIDCF:RaisedFloor	Raised Floor	Raised Floor as classified within Civil and structural.	D2
AIDCF:Roof	Roof	Roof as classified within Civil and structural.	D2
AIDCF:Wall	Wall	Wall as classified within Civil and structural.	D2
AIDCF:Ceiling	Ceiling	Ceiling as classified within Civil and structural.	D2
AIDCF:RackAnchorage	Rack Anchorage	Rack Anchorage as classified within Civil and structural.	D2
AIDCF:SeismicRestraint	Seismic Restraint	Seismic Restraint as classified within Civil and structural.	D2
AIDCF:Load-BearingElement	Load-Bearing Element	Load-Bearing Element as classified within Civil and structural.	D2
AIDCF:IncomingSupply	Incoming Supply	Incoming Supply as classified within Electrical architecture.	D2
AIDCF:Substation	Substation	Substation as classified within Electrical architecture.	D2
AIDCF:Transformer	Transformer	Transformer as classified within Electrical architecture.	D2
AIDCF:Switchgear	Switchgear	Switchgear as classified within Electrical architecture.	D2
AIDCF:ProtectionRelay	Protection Relay	Protection Relay as classified within Electrical architecture.	D2
AIDCF:TransferScheme	Transfer Scheme	Transfer Scheme as classified within Electrical architecture.	D2
AIDCF:EarthingSystem	Earthing System	Earthing System as classified within Electrical architecture.	D2
AIDCF:LightningProtection	Lightning Protection	Lightning Protection as classified within Electrical architecture.	D2
AIDCF:ChilledWaterPlant	Chilled Water Plant	Chilled Water Plant as classified within Mechanical architecture.	D2
AIDCF:CondenserSystem	Condenser System	Condenser System as classified within Mechanical architecture.	D2
AIDCF:PumpSystem	Pump System	Pump System as classified within Mechanical architecture.	D2
AIDCF:AirDistribution	Air Distribution	Air Distribution as classified within Mechanical architecture.	D2
AIDCF:Ventilation	Ventilation	Ventilation as classified within Mechanical architecture.	D2
AIDCF:Pressurisation	Pressurisation	Pressurisation as classified within Mechanical architecture.	D2
AIDCF:Drainage	Drainage	Drainage as classified within Mechanical architecture.	D2
AIDCF:DetectionZone	Detection Zone	Detection Zone as classified within Fire and life safety.	D2

Canonical ID	Preferred label	Definition / use	Domain
AIDCF:FireAlarmPanel	Fire Alarm Panel	Fire Alarm Panel as classified within Fire and life safety.	D2
AIDCF:SuppressionZone	Suppression Zone	Suppression Zone as classified within Fire and life safety.	D2
AIDCF:AgentSystem	Agent System	Agent System as classified within Fire and life safety.	D2
AIDCF:SprinklerSystem	Sprinkler System	Sprinkler System as classified within Fire and life safety.	D2
AIDCF:SmokeControl	Smoke Control	Smoke Control as classified within Fire and life safety.	D2
AIDCF:EmergencyEgress	Emergency Egress	Emergency Egress as classified within Fire and life safety.	D2
AIDCF:EmergencyPowerOff	Emergency Power Off	Emergency Power Off as classified within Fire and life safety.	D2
AIDCF:WaterSource	Water Source	Water Source as classified within Water and leak management.	D2
AIDCF:Pipework	Pipework	Pipework as classified within Water and leak management.	D2
AIDCF:Valve	Valve	Valve as classified within Water and leak management.	D2
AIDCF:Drain	Drain	Drain as classified within Water and leak management.	D2
AIDCF:Sump	Sump	Sump as classified within Water and leak management.	D2
AIDCF:LeakDetectionZone	Leak Detection Zone	Leak Detection Zone as classified within Water and leak management.	D2
AIDCF:WaterTreatmentSystem	Water Treatment System	Water Treatment System as classified within Water and leak management.	D2
AIDCF:BMS	BMS	BMS as classified within Facility control and telemetry.	D2
AIDCF:EPMS	EPMS	EPMS as classified within Facility control and telemetry.	D2
AIDCF:SCADA	SCADA	SCADA as classified within Facility control and telemetry.	D2
AIDCF:PLC	PLC	PLC as classified within Facility control and telemetry.	D2
AIDCF:RTU	RTU	RTU as classified within Facility control and telemetry.	D2
AIDCF:Sensor	Sensor	Sensor as classified within Facility control and telemetry.	D2
AIDCF:Actuator	Actuator	Actuator as classified within Facility control and telemetry.	D2
AIDCF:Gateway	Gateway	Gateway as classified within Facility control and telemetry.	D2
AIDCF:Historian	Historian	Historian as classified within Facility control and telemetry.	D2
AIDCF:AlarmConsole	Alarm Console	Alarm Console as classified within Facility control and telemetry.	D2
AIDCF:DesignBasis	Design Basis	Design Basis as classified within Engineering authority and records.	D2

Canonical ID	Preferred label	Definition / use	Domain
AIDCF:Single-LineDiagram	Single-Line Diagram	Single-Line Diagram as classified within Engineering authority and records.	D2
AIDCF:P&ID	P&ID	P&ID as classified within Engineering authority and records.	D2
AIDCF:As-BuiltDrawing	As-Built Drawing	As-Built Drawing as classified within Engineering authority and records.	D2
AIDCF:EngineeringStudy	Engineering Study	Engineering Study as classified within Engineering authority and records.	D2
AIDCF:CommissioningScript	Commissioning Script	Commissioning Script as classified within Engineering authority and records.	D2
AIDCF:DeviationRecord	Deviation Record	Deviation Record as classified within Engineering authority and records.	D2
AIDCF:UtilityFeed	Utility Feed	Utility Feed as classified within Utility and generation.	D3
AIDCF:GridConnection	Grid Connection	Grid Connection as classified within Utility and generation.	D3
AIDCF:On-SiteGeneration	On-Site Generation	On-Site Generation as classified within Utility and generation.	D3
AIDCF:Generator	Generator	Generator as classified within Utility and generation.	D3
AIDCF:FuelTank	Fuel Tank	Fuel Tank as classified within Utility and generation.	D3
AIDCF:FuelSystem	Fuel System	Fuel System as classified within Utility and generation.	D3
AIDCF:StartingSystem	Starting System	Starting System as classified within Utility and generation.	D3
AIDCF:ExhaustSystem	Exhaust System	Exhaust System as classified within Utility and generation.	D3
AIDCF:Transformer	Transformer	Transformer as classified within Power conversion and distribution.	D3
AIDCF:MVSwitchgear	MV Switchgear	MV Switchgear as classified within Power conversion and distribution.	D3
AIDCF:LVSwitchgear	LV Switchgear	LV Switchgear as classified within Power conversion and distribution.	D3
AIDCF:UPS	UPS	UPS as classified within Power conversion and distribution.	D3
AIDCF:Battery	Battery	Battery as classified within Power conversion and distribution.	D3
AIDCF:EnergyStorage	Energy Storage	Energy Storage as classified within Power conversion and distribution.	D3
AIDCF:StaticTransferSwitch	Static Transfer Switch	Static Transfer Switch as classified within Power conversion and distribution.	D3
AIDCF:Busway	Busway	Busway as classified within Power conversion and distribution.	D3
AIDCF:PDU	PDU	PDU as classified within Power conversion and distribution.	D3
AIDCF:RPP	RPP	RPP as classified within Power conversion and distribution.	D3
AIDCF:RackPDU	Rack PDU	Rack PDU as classified within Power conversion and distribution.	D3

Canonical ID	Preferred label	Definition / use	Domain
AIDCF:PowerSupplyUnit	Power Supply Unit	Power Supply Unit as classified within Power conversion and distribution.	D3
AIDCF:ProtectionDevice	Protection Device	Protection Device as classified within Power quality and protection.	D3
AIDCF:Relay	Relay	Relay as classified within Power quality and protection.	D3
AIDCF:Meter	Meter	Meter as classified within Power quality and protection.	D3
AIDCF:HarmonicFilter	Harmonic Filter	Harmonic Filter as classified within Power quality and protection.	D3
AIDCF:SurgeProtection	Surge Protection	Surge Protection as classified within Power quality and protection.	D3
AIDCF:Grounding	Grounding	Grounding as classified within Power quality and protection.	D3
AIDCF:PowerQualityMonitor	Power Quality Monitor	Power Quality Monitor as classified within Power quality and protection.	D3
AIDCF:HeatSink	Heat Sink	Heat Sink as classified within Heat capture.	D3
AIDCF:ColdPlate	Cold Plate	Cold Plate as classified within Heat capture.	D3
AIDCF:Rear-DoorHeatExchanger	Rear-Door Heat Exchanger	Rear-Door Heat Exchanger as classified within Heat capture.	D3
AIDCF:AirHandler	Air Handler	Air Handler as classified within Heat capture.	D3
AIDCF:In-RowCooler	In-Row Cooler	In-Row Cooler as classified within Heat capture.	D3
AIDCF:ImmersionTank	Immersion Tank	Immersion Tank as classified within Heat capture.	D3
AIDCF:CDU	CDU	CDU as classified within Coolant distribution.	D3
AIDCF:PrimaryLoop	Primary Loop	Primary Loop as classified within Coolant distribution.	D3
AIDCF:SecondaryLoop	Secondary Loop	Secondary Loop as classified within Coolant distribution.	D3
AIDCF:Pump	Pump	Pump as classified within Coolant distribution.	D3
AIDCF:Manifold	Manifold	Manifold as classified within Coolant distribution.	D3
AIDCF:QuickDisconnect	Quick Disconnect	Quick Disconnect as classified within Coolant distribution.	D3
AIDCF:Hose	Hose	Hose as classified within Coolant distribution.	D3
AIDCF:Pipe	Pipe	Pipe as classified within Coolant distribution.	D3
AIDCF:HeatExchanger	Heat Exchanger	Heat Exchanger as classified within Coolant distribution.	D3
AIDCF:Chiller	Chiller	Chiller as classified within Heat rejection.	D3
AIDCF:CoolingTower	Cooling Tower	Cooling Tower as classified within Heat rejection.	D3

Canonical ID	Preferred label	Definition / use	Domain
AIDCF:DryCooler	Dry Cooler	Dry Cooler as classified within Heat rejection.	D3
AIDCF:Condenser	Condenser	Condenser as classified within Heat rejection.	D3
AIDCF:EvaporativeSystem	Evaporative System	Evaporative System as classified within Heat rejection.	D3
AIDCF:Heat-ReuseInterface	Heat-Reuse Interface	Heat-Reuse Interface as classified within Heat rejection.	D3
AIDCF:TemperatureSensor	Temperature Sensor	Temperature Sensor as classified within Thermal and leak monitoring.	D3
AIDCF:HumiditySensor	Humidity Sensor	Humidity Sensor as classified within Thermal and leak monitoring.	D3
AIDCF:FlowSensor	Flow Sensor	Flow Sensor as classified within Thermal and leak monitoring.	D3
AIDCF:PressureSensor	Pressure Sensor	Pressure Sensor as classified within Thermal and leak monitoring.	D3
AIDCF:LeakSensor	Leak Sensor	Leak Sensor as classified within Thermal and leak monitoring.	D3
AIDCF:CoolantQualitySensor	Coolant Quality Sensor	Coolant Quality Sensor as classified within Thermal and leak monitoring.	D3
AIDCF:ThermalCamera	Thermal Camera	Thermal Camera as classified within Thermal and leak monitoring.	D3
AIDCF:GridElectricity	Grid Electricity	Grid Electricity as classified within Energy resources.	D4
AIDCF:RenewableElectricity	Renewable Electricity	Renewable Electricity as classified within Energy resources.	D4
AIDCF:On-SiteEnergy	On-Site Energy	On-Site Energy as classified within Energy resources.	D4
AIDCF:StoredEnergy	Stored Energy	Stored Energy as classified within Energy resources.	D4
AIDCF:Fuel	Fuel	Fuel as classified within Energy resources.	D4
AIDCF:PotableWater	Potable Water	Potable Water as classified within Water resources.	D4
AIDCF:ProcessWater	Process Water	Process Water as classified within Water resources.	D4
AIDCF:ReclaimedWater	Reclaimed Water	Reclaimed Water as classified within Water resources.	D4
AIDCF:CoolingWater	Cooling Water	Cooling Water as classified within Water resources.	D4
AIDCF:DischargeWater	Discharge Water	Discharge Water as classified within Water resources.	D4
AIDCF:AmbientTemperature	Ambient Temperature	Ambient Temperature as classified within Environmental conditions.	D4
AIDCF:Humidity	Humidity	Humidity as classified within Environmental conditions.	D4
AIDCF:AirQuality	Air Quality	Air Quality as classified within Environmental conditions.	D4
AIDCF:ParticulateLevel	Particulate Level	Particulate Level as classified within Environmental conditions.	D4

Canonical ID	Preferred label	Definition / use	Domain
AIDCF:Noise	Noise	Noise as classified within Environmental conditions.	D4
AIDCF:Vibration	Vibration	Vibration as classified within Environmental conditions.	D4
AIDCF:FloodExposure	Flood Exposure	Flood Exposure as classified within Environmental conditions.	D4
AIDCF:HeatStress	Heat Stress	Heat Stress as classified within Environmental conditions.	D4
AIDCF:Refrigerant	Refrigerant	Refrigerant as classified within Material and chemical resources.	D4
AIDCF:Coolant	Coolant	Coolant as classified within Material and chemical resources.	D4
AIDCF:FireSuppressant	Fire Suppressant	Fire Suppressant as classified within Material and chemical resources.	D4
AIDCF:BatteryChemistry	Battery Chemistry	Battery Chemistry as classified within Material and chemical resources.	D4
AIDCF:HazardousMaterial	Hazardous Material	Hazardous Material as classified within Material and chemical resources.	D4
AIDCF:PUERecord	PUE Record	PUE Record as classified within Performance indicators.	D4
AIDCF:WUERecord	WUE Record	WUE Record as classified within Performance indicators.	D4
AIDCF:CUERecord	CUE Record	CUE Record as classified within Performance indicators.	D4
AIDCF:EnergyReuseRecord	Energy Reuse Record	Energy Reuse Record as classified within Performance indicators.	D4
AIDCF:CoolingEfficiencyRecord	Cooling Efficiency Record	Cooling Efficiency Record as classified within Performance indicators.	D4
AIDCF:ComputeUtilisationRecord	Compute Utilisation Record	Compute Utilisation Record as classified within Performance indicators.	D4
AIDCF:Permit	Permit	Permit as classified within Environmental obligations.	D4
AIDCF:ReportingRequirement	Reporting Requirement	Reporting Requirement as classified within Environmental obligations.	D4
AIDCF:EmissionLimit	Emission Limit	Emission Limit as classified within Environmental obligations.	D4
AIDCF:WaterLimit	Water Limit	Water Limit as classified within Environmental obligations.	D4
AIDCF:WasteRequirement	Waste Requirement	Waste Requirement as classified within Environmental obligations.	D4
AIDCF:EnvironmentalCommitment	Environmental Commitment	Environmental Commitment as classified within Environmental obligations.	D4
AIDCF:OperatingModel	Operating Model	Operating Model as classified within Governance and authority.	D5
AIDCF:Role	Role	Role as classified within Governance and authority.	D5
AIDCF:Delegation	Delegation	Delegation as classified within Governance and authority.	D5
AIDCF:ApprovalAuthority	Approval Authority	Approval Authority as classified within Governance and authority.	D5

Canonical ID	Preferred label	Definition / use	Domain
AIDCF:EscalationPath	Escalation Path	Escalation Path as classified within Governance and authority.	D5
AIDCF:RiskAcceptance	Risk Acceptance	Risk Acceptance as classified within Governance and authority.	D5
AIDCF:Procedure	Procedure	Procedure as classified within Work control.	D5
AIDCF:MethodStatement	Method Statement	Method Statement as classified within Work control.	D5
AIDCF:PermittoWork	Permit to Work	Permit to Work as classified within Work control.	D5
AIDCF:LockoutTagout	Lockout Tagout	Lockout Tagout as classified within Work control.	D5
AIDCF:MaintenanceWindow	Maintenance Window	Maintenance Window as classified within Work control.	D5
AIDCF:ChangeRecord	Change Record	Change Record as classified within Work control.	D5
AIDCF:TemporaryConfiguration	Temporary Configuration	Temporary Configuration as classified within Work control.	D5
AIDCF:MaintenancePlan	Maintenance Plan	Maintenance Plan as classified within Maintenance and reliability.	D5
AIDCF:WorkOrder	Work Order	Work Order as classified within Maintenance and reliability.	D5
AIDCF:ConditionMonitoring	Condition Monitoring	Condition Monitoring as classified within Maintenance and reliability.	D5
AIDCF:Inspection	Inspection	Inspection as classified within Maintenance and reliability.	D5
AIDCF:Calibration	Calibration	Calibration as classified within Maintenance and reliability.	D5
AIDCF:SparePart	Spare Part	Spare Part as classified within Maintenance and reliability.	D5
AIDCF:ObsolescenceRecord	Obsolescence Record	Obsolescence Record as classified within Maintenance and reliability.	D5
AIDCF:Shift	Shift	Shift as classified within Staffing and competence.	D5
AIDCF:Roster	Roster	Roster as classified within Staffing and competence.	D5
AIDCF:CompetencyRecord	Competency Record	Competency Record as classified within Staffing and competence.	D5
AIDCF:Authorisation	Authorisation	Authorisation as classified within Staffing and competence.	D5
AIDCF:TrainingRecord	Training Record	Training Record as classified within Staffing and competence.	D5
AIDCF:Contractor	Contractor	Contractor as classified within Staffing and competence.	D5
AIDCF:SubjectMatterExpert	Subject Matter Expert	Subject Matter Expert as classified within Staffing and competence.	D5
AIDCF:Alarm	Alarm	Alarm as classified within Incident and emergency.	D5
AIDCF:Incident	Incident	Incident as classified within Incident and emergency.	D5

Canonical ID	Preferred label	Definition / use	Domain
AIDCF:EventLog	Event Log	Event Log as classified within Incident and emergency.	D5
AIDCF:EmergencyProcedure	Emergency Procedure	Emergency Procedure as classified within Incident and emergency.	D5
AIDCF:CrisisTeam	Crisis Team	Crisis Team as classified within Incident and emergency.	D5
AIDCF:Exercise	Exercise	Exercise as classified within Incident and emergency.	D5
AIDCF:Post-IncidentReview	Post-Incident Review	Post-Incident Review as classified within Incident and emergency.	D5
AIDCF:Supplier	Supplier	Supplier as classified within Supplier and service management.	D5
AIDCF:MaintenanceContract	Maintenance Contract	Maintenance Contract as classified within Supplier and service management.	D5
AIDCF:ServiceLevel	Service Level	Service Level as classified within Supplier and service management.	D5
AIDCF:RemoteSupport	Remote Support	Remote Support as classified within Supplier and service management.	D5
AIDCF:EscalationContact	Escalation Contact	Escalation Contact as classified within Supplier and service management.	D5
AIDCF:LeadTime	Lead Time	Lead Time as classified within Supplier and service management.	D5
AIDCF:SupplyConstraint	Supply Constraint	Supply Constraint as classified within Supplier and service management.	D5
AIDCF:ConfigurationBaseline	Configuration Baseline	Configuration Baseline as classified within Configuration and documentation.	D5
AIDCF:AssetRegister	Asset Register	Asset Register as classified within Configuration and documentation.	D5
AIDCF:DrawingRegister	Drawing Register	Drawing Register as classified within Configuration and documentation.	D5
AIDCF:ProcedureRegister	Procedure Register	Procedure Register as classified within Configuration and documentation.	D5
AIDCF:Revision	Revision	Revision as classified within Configuration and documentation.	D5
AIDCF:DocumentCustodian	Document Custodian	Document Custodian as classified within Configuration and documentation.	D5
AIDCF:Accelerator	Accelerator	Accelerator as classified within Accelerated compute.	D6
AIDCF:GPU	GPU	GPU as classified within Accelerated compute.	D6
AIDCF:AIASIC	AI ASIC	AI ASIC as classified within Accelerated compute.	D6
AIDCF:ComputeNode	Compute Node	Compute Node as classified within Accelerated compute.	D6
AIDCF:GPUBaseboard	GPU Baseboard	GPU Baseboard as classified within Accelerated compute.	D6
AIDCF:CPU	CPU	CPU as classified within Accelerated compute.	D6
AIDCF:Memory	Memory	Memory as classified within Accelerated compute.	D6

Canonical ID	Preferred label	Definition / use	Domain
AIDCF:DPU	DPU	DPU as classified within Accelerated compute.	D6
AIDCF:NIC	NIC	NIC as classified within Accelerated compute.	D6
AIDCF:ComputeTray	Compute Tray	Compute Tray as classified within Rack-scale systems.	D6
AIDCF:Rack-ScaleSystem	Rack-Scale System	Rack-Scale System as classified within Rack-scale systems.	D6
AIDCF:Scale-UpDomain	Scale-Up Domain	Scale-Up Domain as classified within Rack-scale systems.	D6
AIDCF:NVLinkDomain	NVLink Domain	NVLink Domain as classified within Rack-scale systems.	D6
AIDCF:Enclosure	Enclosure	Enclosure as classified within Rack-scale systems.	D6
AIDCF:Backplane	Backplane	Backplane as classified within Rack-scale systems.	D6
AIDCF:PowerShelf	Power Shelf	Power Shelf as classified within Rack-scale systems.	D6
AIDCF:RackController	Rack Controller	Rack Controller as classified within Rack-scale systems.	D6
AIDCF:ComputeCluster	Compute Cluster	Compute Cluster as classified within Cluster systems.	D6
AIDCF:ClusterPartition	Cluster Partition	Cluster Partition as classified within Cluster systems.	D6
AIDCF:NodePool	Node Pool	Node Pool as classified within Cluster systems.	D6
AIDCF:HeadNode	Head Node	Head Node as classified within Cluster systems.	D6
AIDCF:LoginNode	Login Node	Login Node as classified within Cluster systems.	D6
AIDCF:ManagementNode	Management Node	Management Node as classified within Cluster systems.	D6
AIDCF:ProvisioningNode	Provisioning Node	Provisioning Node as classified within Cluster systems.	D6
AIDCF:BMC	BMC	BMC as classified within Management and firmware.	D6
AIDCF:Firmware	Firmware	Firmware as classified within Management and firmware.	D6
AIDCF:BIOS	BIOS	BIOS as classified within Management and firmware.	D6
AIDCF:Driver	Driver	Driver as classified within Management and firmware.	D6
AIDCF:FabricManager	Fabric Manager	Fabric Manager as classified within Management and firmware.	D6
AIDCF:ClusterManager	Cluster Manager	Cluster Manager as classified within Management and firmware.	D6
AIDCF:ProvisioningService	Provisioning Service	Provisioning Service as classified within Management and firmware.	D6
AIDCF:Scheduler	Scheduler	Scheduler as classified within Management and firmware.	D6

Canonical ID	Preferred label	Definition / use	Domain
AIDCF:Orchestrator	Orchestrator	Orchestrator as classified within Management and firmware.	D6
AIDCF:Queue	Queue	Queue as classified within Workload infrastructure.	D6
AIDCF:Partition	Partition	Partition as classified within Workload infrastructure.	D6
AIDCF:Reservation	Reservation	Reservation as classified within Workload infrastructure.	D6
AIDCF:ContainerRuntime	Container Runtime	Container Runtime as classified within Workload infrastructure.	D6
AIDCF:KubernetesCluster	Kubernetes Cluster	Kubernetes Cluster as classified within Workload infrastructure.	D6
AIDCF:JobScheduler	Job Scheduler	Job Scheduler as classified within Workload infrastructure.	D6
AIDCF:CheckpointAgent	Checkpoint Agent	Checkpoint Agent as classified within Workload infrastructure.	D6
AIDCF:GPUTelemetry	GPU Telemetry	GPU Telemetry as classified within Compute observability.	D6
AIDCF:NodeTelemetry	Node Telemetry	Node Telemetry as classified within Compute observability.	D6
AIDCF:HealthService	Health Service	Health Service as classified within Compute observability.	D6
AIDCF:EventLog	Event Log	Event Log as classified within Compute observability.	D6
AIDCF:PerformanceCounter	Performance Counter	Performance Counter as classified within Compute observability.	D6
AIDCF:ErrorCounter	Error Counter	Error Counter as classified within Compute observability.	D6
AIDCF:GoldenImage	Golden Image	Golden Image as classified within Compute lifecycle.	D6
AIDCF:FirmwareBaseline	Firmware Baseline	Firmware Baseline as classified within Compute lifecycle.	D6
AIDCF:CompatibilityMatrix	Compatibility Matrix	Compatibility Matrix as classified within Compute lifecycle.	D6
AIDCF:HardwareSupportContract	Hardware Support Contract	Hardware Support Contract as classified within Compute lifecycle.	D6
AIDCF:RMAProcess	RMA Process	RMA Process as classified within Compute lifecycle.	D6
AIDCF:SecureDecommissioningRecord	Secure Decommissioning Record	Secure Decommissioning Record as classified within Compute lifecycle.	D6
AIDCF:ParallelFileSystem	Parallel File System	Parallel File System as classified within Storage systems.	D7
AIDCF:Scale-OutNAS	Scale-Out NAS	Scale-Out NAS as classified within Storage systems.	D7
AIDCF:ObjectStorage	Object Storage	Object Storage as classified within Storage systems.	D7
AIDCF:BlockStorage	Block Storage	Block Storage as classified within Storage systems.	D7
AIDCF:LocalNVMe	Local NVMe	Local NVMe as classified within Storage systems.	D7

Canonical ID	Preferred label	Definition / use	Domain
AIDCF:BurstBuffer	Burst Buffer	Burst Buffer as classified within Storage systems.	D7
AIDCF:ArchiveStorage	Archive Storage	Archive Storage as classified within Storage systems.	D7
AIDCF:StorageNode	Storage Node	Storage Node as classified within Storage nodes and media.	D7
AIDCF:MetadataServer	Metadata Server	Metadata Server as classified within Storage nodes and media.	D7
AIDCF:ObjectStorageNode	Object Storage Node	Object Storage Node as classified within Storage nodes and media.	D7
AIDCF:Controller	Controller	Controller as classified within Storage nodes and media.	D7
AIDCF:SSD	SSD	SSD as classified within Storage nodes and media.	D7
AIDCF:HDD	HDD	HDD as classified within Storage nodes and media.	D7
AIDCF:Tape	Tape	Tape as classified within Storage nodes and media.	D7
AIDCF:NVRAM	NVRAM	NVRAM as classified within Storage nodes and media.	D7
AIDCF:Namespace	Namespace	Namespace as classified within Data-path services.	D7
AIDCF:MetadataService	Metadata Service	Metadata Service as classified within Data-path services.	D7
AIDCF:DataMover	Data Mover	Data Mover as classified within Data-path services.	D7
AIDCF:Cache	Cache	Cache as classified within Data-path services.	D7
AIDCF:Gateway	Gateway	Gateway as classified within Data-path services.	D7
AIDCF:ClientDriver	Client Driver	Client Driver as classified within Data-path services.	D7
AIDCF:MountService	Mount Service	Mount Service as classified within Data-path services.	D7
AIDCF:DatasetRepository	Dataset Repository	Dataset Repository as classified within AI artifact services.	D7
AIDCF:ModelRepository	Model Repository	Model Repository as classified within AI artifact services.	D7
AIDCF:CheckpointRepository	Checkpoint Repository	Checkpoint Repository as classified within AI artifact services.	D7
AIDCF:FeatureStore	Feature Store	Feature Store as classified within AI artifact services.	D7
AIDCF:ArtifactRegistry	Artifact Registry	Artifact Registry as classified within AI artifact services.	D7
AIDCF:Snapshot	Snapshot	Snapshot as classified within Protection and recovery.	D7
AIDCF:Replica	Replica	Replica as classified within Protection and recovery.	D7
AIDCF:BackupCopy	Backup Copy	Backup Copy as classified within Protection and recovery.	D7

Canonical ID	Preferred label	Definition / use	Domain
AIDCF:ErasureCodingGroup	Erasure Coding Group	Erasure Coding Group as classified within Protection and recovery.	D7
AIDCF:RecoveryCopy	Recovery Copy	Recovery Copy as classified within Protection and recovery.	D7
AIDCF:ImmutableCopy	Immutable Copy	Immutable Copy as classified within Protection and recovery.	D7
AIDCF:StorageManager	Storage Manager	Storage Manager as classified within Storage management.	D7
AIDCF:CapacityManager	Capacity Manager	Capacity Manager as classified within Storage management.	D7
AIDCF:TieringPolicy	Tiering Policy	Tiering Policy as classified within Storage management.	D7
AIDCF:RetentionPolicy	Retention Policy	Retention Policy as classified within Storage management.	D7
AIDCF:IntegrityScanner	Integrity Scanner	Integrity Scanner as classified within Storage management.	D7
AIDCF:TelemetryService	Telemetry Service	Telemetry Service as classified within Storage management.	D7
AIDCF:IngestionInterface	Ingestion Interface	Ingestion Interface as classified within Data logistics interfaces.	D7
AIDCF:ExportInterface	Export Interface	Export Interface as classified within Data logistics interfaces.	D7
AIDCF:DataTransferNode	Data Transfer Node	Data Transfer Node as classified within Data logistics interfaces.	D7
AIDCF:WANTransferService	WAN Transfer Service	WAN Transfer Service as classified within Data logistics interfaces.	D7
AIDCF:StagingArea	Staging Area	Staging Area as classified within Data logistics interfaces.	D7
AIDCF:Scale-UpFabric	Scale-Up Fabric	Scale-Up Fabric as classified within Scale-up interconnect.	D8
AIDCF:SwitchFabric	Switch Fabric	Switch Fabric as classified within Scale-up interconnect.	D8
AIDCF:InterconnectLink	Interconnect Link	Interconnect Link as classified within Scale-up interconnect.	D8
AIDCF:TopologyDomain	Topology Domain	Topology Domain as classified within Scale-up interconnect.	D8
AIDCF:InfiniBandFabric	InfiniBand Fabric	InfiniBand Fabric as classified within Scale-out compute fabric.	D8
AIDCF:EthernetAIFabric	Ethernet AI Fabric	Ethernet AI Fabric as classified within Scale-out compute fabric.	D8
AIDCF:RoCEFabric	RoCE Fabric	RoCE Fabric as classified within Scale-out compute fabric.	D8
AIDCF:ComputeLeaf	Compute Leaf	Compute Leaf as classified within Scale-out compute fabric.	D8
AIDCF:ComputeSpine	Compute Spine	Compute Spine as classified within Scale-out compute fabric.	D8
AIDCF:FabricManager	Fabric Manager	Fabric Manager as classified within Scale-out compute fabric.	D8
AIDCF:StorageNetwork	Storage Network	Storage Network as classified within Storage fabric.	D8

Canonical ID	Preferred label	Definition / use	Domain
AIDCF:StorageLeaf	Storage Leaf	Storage Leaf as classified within Storage fabric.	D8
AIDCF:StorageSpine	Storage Spine	Storage Spine as classified within Storage fabric.	D8
AIDCF:SANFabric	SAN Fabric	SAN Fabric as classified within Storage fabric.	D8
AIDCF:StorageGateway	Storage Gateway	Storage Gateway as classified within Storage fabric.	D8
AIDCF:FrontendNetwork	Frontend Network	Frontend Network as classified within Frontend and service network.	D8
AIDCF:LoadBalancer	Load Balancer	Load Balancer as classified within Frontend and service network.	D8
AIDCF:ServiceGateway	Service Gateway	Service Gateway as classified within Frontend and service network.	D8
AIDCF:EdgeRouter	Edge Router	Edge Router as classified within Frontend and service network.	D8
AIDCF:Firewall	Firewall	Firewall as classified within Frontend and service network.	D8
AIDCF:APINetwork	API Network	API Network as classified within Frontend and service network.	D8
AIDCF:ManagementNetwork	Management Network	Management Network as classified within Management and control network.	D8
AIDCF:Out-of-BandNetwork	Out-of-Band Network	Out-of-Band Network as classified within Management and control network.	D8
AIDCF:ConsoleServer	Console Server	Console Server as classified within Management and control network.	D8
AIDCF:ManagementSwitch	Management Switch	Management Switch as classified within Management and control network.	D8
AIDCF:JumpService	Jump Service	Jump Service as classified within Management and control network.	D8
AIDCF:TimeService	Time Service	Time Service as classified within Timing and core services.	D8
AIDCF:DNSService	DNS Service	DNS Service as classified within Timing and core services.	D8
AIDCF:DHCPService	DHCP Service	DHCP Service as classified within Timing and core services.	D8
AIDCF:IPAMService	IPAM Service	IPAM Service as classified within Timing and core services.	D8
AIDCF:CertificateService	Certificate Service	Certificate Service as classified within Timing and core services.	D8
AIDCF:AuthenticationService	Authentication Service	Authentication Service as classified within Timing and core services.	D8
AIDCF:Cable	Cable	Cable as classified within Physical connectivity.	D8
AIDCF:OpticalFibre	Optical Fibre	Optical Fibre as classified within Physical connectivity.	D8
AIDCF:CopperLink	Copper Link	Copper Link as classified within Physical connectivity.	D8
AIDCF:Transceiver	Transceiver	Transceiver as classified within Physical connectivity.	D8

Canonical ID	Preferred label	Definition / use	Domain
AIDCF:PatchPanel	Patch Panel	Patch Panel as classified within Physical connectivity.	D8
AIDCF:Meet-MeRoom	Meet-Me Room	Meet-Me Room as classified within Physical connectivity.	D8
AIDCF:Cross-Connect	Cross-Connect	Cross-Connect as classified within Physical connectivity.	D8
AIDCF:FlowTelemetry	Flow Telemetry	Flow Telemetry as classified within Network observability.	D8
AIDCF:FabricTelemetry	Fabric Telemetry	Fabric Telemetry as classified within Network observability.	D8
AIDCF:PacketCaptureService	Packet Capture Service	Packet Capture Service as classified within Network observability.	D8
AIDCF:NetworkController	Network Controller	Network Controller as classified within Network observability.	D8
AIDCF:CongestionMonitor	Congestion Monitor	Congestion Monitor as classified within Network observability.	D8
AIDCF:PathMonitor	Path Monitor	Path Monitor as classified within Network observability.	D8

Appendix B - Canonical relationship registry

Predicate	Subject	Object	Logic	Inverse	Meaning
PART_OF	Entity	Entity	Transitive	HAS_PART	Composition.
CONTAINS	Location/System	Entity	Transitive	LOCATED_IN/PART_OF	Containment.
LOCATED_IN	Physical Entity	Location	Transitive	CONTAINS	Physical placement.
MEMBER_OF	Entity	Group	No	HAS_MEMBER	Group membership.
IMPLEMENTS	Asset/System	Function	No	IMPLEMENTED_BY	Realises a required function.
SUPPORTS	Entity	Capability/Outcome	No	SUPPORTED_BY	Contributes to outcome.
INTERFACES_WITH	Entity	Entity	Symmetric	INTERFACES_WITH	Defined exchange boundary.
CONNECTED_TO	Entity	Entity	Symmetric unless directed	CONNECTED_TO	Realised connection.
ROUTES_THROUGH	Flow/Service	Entity	No	CARRIES	Path relationship.
DEPENDS_ON	Consumer entity	Provider/entity/condition	Directional	Declared where applicable	General reliance.
POWERED_BY	Load/system	Electrical source/path	Directional	Declared where applicable	Electrical reliance.
COOLED_BY	Heat source/system	Cooling system/path	Directional	Declared where applicable	Thermal reliance.
SUPPLIED_BY	Consumer	Provider/resource	Directional	Declared where applicable	Resource or service supply.
CONTROLLED_BY	Asset/system	Controller/actor	Directional	Declared where applicable	Operational or automated control.
MONITORED_BY	Entity	Monitoring entity	Directional	Declared where applicable	Visibility and detection reliance.
PROTECTED_BY	Entity	Safeguard	Directional	Declared where applicable	Protective relationship.
AUTHENTICATED_BY	Entity/service	Identity service	Directional	Declared where applicable	Identity reliance.

Predicate	Subject	Object	Logic	Inverse	Meaning
TIMED_BY	Entity/service	Time source	Directional	Declared where applicable	Timing reliance.
CONFIGURED_BY	Entity	Configuration source/actor	Directional	Declared where applicable	Configuration reliance.
RECOVERED_BY	Entity/capability	Recovery asset/process	Directional	Declared where applicable	Restoration reliance.
MAINTAINED_BY	Entity	Actor/organisation	Directional	Declared where applicable	Maintenance reliance.
OPERATED_BY	Entity/system	Actor/organisation	Directional	Declared where applicable	Operational responsibility.
EVIDENCED_BY	Assertion/entity	Evidence object	Directional	Declared where applicable	Evidence support.
TESTED_BY	Assertion/safeguard	Test record	Directional	Declared where applicable	Validation link.
SHARES_DEPENDENCY_WITH	Entity	Entity	Predicate-specific	May be symmetric	Common provider exposure.
IN_SAME_FAULT_DOMAIN_AS	Entity	Entity	Predicate-specific	May be symmetric	Shared failure exposure.
DIVERSE_FROM	Entity	Entity	Predicate-specific	May be symmetric	Declared material difference.
INDEPENDENT_OF	Entity	Entity	Predicate-specific	May be symmetric	Evidence-bounded relevant independence.
REDUNDANT_WITH	Entity	Entity	Predicate-specific	May be symmetric	Alternative for a protected function.
CONCENTRATES	Entity	Entity	Predicate-specific	May be symmetric	Aggregates capability or consequence.
SINGLE_POINT_FOR	Entity	Entity	Predicate-specific	May be symmetric	Unmitigated single-point relationship.
ISOLATED_FROM	Entity	Entity	Predicate-specific	May be symmetric	Declared separation.

Appendix C - Minimum property schemas

C.1 Capability Instance

Property	Cardinality	Type
capabilityInstanceId	1	Identifier
name	1	String
capabilityType	1	Controlled term
owner	1	Party
lifecycleStage	1	Enumeration
serviceOutcomes	1..* / 0..*	ServiceOutcome
criticality	1..* / 0..*	Criticality
assessmentBoundary	1	Boundary
operatingEnvelope	1	Envelope
architectureBaseline	1..* / 0..*	Document/Configuration
effectivePeriod	1..* / 0..*	Temporal interval

Property	Cardinality	Type
confidenceState	1	Confidence

C.2 Dependency

Property	Cardinality	Type
dependencyId	1	Identifier
consumer	1	Entity
provider	1	Entity or UNKNOWN
dependencyType	1	Controlled term
strength	1	Enumeration
applicableStates	1..*	OperatingState
capacityRequirement	0..*	CapacityAssertion
substitutability	0..1	Enumeration
owner	1	Party or UNKNOWN
evidence	0..*	EvidenceObject
confidence	1	ConfidenceState
validity	1	Temporal interval

C.3 Evidence Object

Property	Cardinality	Type
evidenceId	1	Identifier
title	1	String
evidenceType	1	Controlled term
source	1	Source
custodian	1	Party
collectionMethod	1	Controlled term
observationTime	1	Date/time
effectivePeriod	0..1	Temporal interval
scope	1	Entity set
integrityStatus	0..1	Controlled term
supportsOrChallenges	1..*	Assertion
limitations	0..*	String
reviewer	0..1	Party

Property	Cardinality	Type
confidence	1	Confidence state

Appendix D - Naming, identifier and namespace rules

- Use singular nouns for classes and active predicates for relationships.
- Avoid acronyms as canonical labels unless universally necessary; retain aliases separately.
- Record regional spellings as labels without creating duplicate semantic classes.
- Keep metrics and units separate from values.
- Do not encode confidentiality, location or owner in public identifiers.
- Use explicit version metadata for vocabularies, schemas and profiles.
- Mark deprecated terms with replacement mappings and retirement rationale.
- Maintain a collision register for extensions and imported vocabularies.

Appendix E - Standards and reference-source map

The sources below informed scope comparisons and terminology analysis. AIDCF does not reproduce proprietary requirements and does not claim conformity with a referenced standard merely by mapping concepts.

Reference	Publisher/type	Relevance
AIDCF Manifesto v1.0	Internal canonical source	Purpose, doctrine, eight domains, evidence and UNKNOWN principles.
AIDCF Core Conceptual Model v1.0	Internal canonical source	Capability, assets, dependencies, capacity, failure, safeguard and assurance concepts.
ISO/IEC 22237-1:2021	ISO	General data-centre concepts, reference models and classification context; the standard explicitly excludes selection/configuration of IT and network equipment.
ISO/IEC 22237 series	ISO/IEC	Facilities and infrastructure topics spanning building, power, environmental control, cabling, security and operations.
EN 50600 series	CENELEC	European data-centre facilities and infrastructure planning, construction and operation context.
NVIDIA Enterprise Reference Architectures	NVIDIA	Validated and repeatable AI-factory designs integrating compute, networking, storage and software.
Open Systems for AI: Blueprint for Scalable Infrastructure	Open Compute Project	Open and interoperable AI infrastructure ecosystem context.
AI-Ready Data Storage Infrastructure taxonomy	IDC	Storage-specific AI readiness and data logistics taxonomy context.
Cisco AI-ready infrastructure design guides	Cisco	Examples of AI infrastructure design profiles and validated designs.

E.1 Public links

ISO/IEC 22237-1:2021: https://www.iso.org/standard/78550.html

NVIDIA Reference Architectures: https://docs.nvidia.com/reference-architectures/index.html

NVIDIA Enterprise Reference Architectures: https://www.nvidia.com/en-us/technologies/enterprise-reference-architecture/

Open Systems for AI whitepaper: https://www.opencompute.org/documents/ocp-open-systems-for-ai-whitepaper-v1-0-0-final-pdf/

Cisco AI-ready infrastructure design guides: https://www.cisco.com/c/en/us/solutions/design-zone/ai-ready-infrastructure.html

Appendix F - Review and acceptance record

Review role	Status	Acceptance focus
Methodology author	Draft completed	Internal coherence and alignment with Artifacts 1 and 2.
Ontology/knowledge-model reviewer	Pending	Class, relationship, constraint and machine-readable integrity.
Data-centre architecture reviewer	Pending	Facility and system taxonomy usability.
Electrical engineering reviewer	Pending	Power semantics and dependency accuracy.
Mechanical/cooling reviewer	Pending	Thermal, hydraulic and liquid-cooling semantics.
Fire/life-safety reviewer	Pending	Boundary and safety terminology.
Physical-security reviewer	Pending	Physical control and zoning taxonomy.
AI compute/HPC reviewer	Pending	Compute, rack-scale and cluster taxonomy.
Storage reviewer	Pending	Data infrastructure and recovery semantics.
Network/interconnect reviewer	Pending	Scale-up, scale-out and management fabrics.
OT/security reviewer	Pending	BMS, EPMS, SCADA and control relationships.
Assurance/evidence reviewer	Pending	UNKNOWN, confidence, provenance and claims.
AI Trust Graph integration reviewer	Pending	Handoff and shared-node semantics.
Legal/IP/trademark review	Pending	Publication claims, licensing and ownership.
Public release	Not approved	All mandatory review gates closed.

F.1 Acceptance criteria

- Canonical terms remain consistent with the Manifesto and Core Conceptual Model.
- All eight domains are represented without becoming isolated silos.
- Capability, outcome, function, system, asset and dependency distinctions are operationally usable.
- Capacity and redundancy claims are condition- and evidence-bounded.
- Time, state, responsibility and UNKNOWN are first-class semantics.
- Artifact 4 can attach threats, hazards, failures and propagation paths without restructuring the ontology.
- Machine-readable schemas can be generated from the normative registries.
- No content creates an unsupported certification or engineering claim.
- Independent multidisciplinary review and change control are recorded.

Change record

Version	Status	Summary
0.1	Architecture draft	Established upper ontology and eight-domain taxonomy.

Version	Status	Summary
0.5	Integrated draft	Added relationships, state, capacity, responsibility, validation and profiles.
1.0	Controlled-review draft	Publication-grade baseline for expert review and machine-readable implementation.

Final doctrine

AIDCF infrastructure is represented only when its meaning can be connected to a Capability Instance, Service Outcome, required function, dependency, state or assurance purpose. Classification alone is insufficient. The ontology must reveal how the system is assembled, what it relies upon, under which conditions claims apply, who is responsible, what evidence supports the representation and what remains UNKNOWN.